



MANAGING WASTE IN FOOD SUPPLY CHAINS WITH SUPPLY CHAIN COLLABORATION AND INTEGRATION

Lappeenranta–Lahti University of Technology LUT

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ABSTRACT

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Managing waste in food supply chains with supply chain collaboration and integration

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Food supply chains are under increasing pressure to remain efficient and competitive in a market impacted by various geopolitical, economic, and climatic factors. As a result, companies are exploring ways to improve their operations and performance. One approach that food supply chain members use to achieve these goals is collaborative methods for managing food loss and waste.

The objective of this study was to evaluate the use of supply chain collaboration and integration in managing food loss and waste in food supply chains. Specifically, the study investigated how collaboration and integration between retailers and suppliers can be utilized to manage and reduce food loss and waste in the food supply chain.

The study was conducted as a qualitative multiple-case study, using research data gathered through semi-structured interviews with six food retailing companies operating in the Finnish food industry. A content analysis method was utilized to analyse the research data.

The empirical findings suggest that supply chain collaboration and integration are valuable and beneficial elements in managing food loss and waste in food supply chains. Supply chain members are interested in collaboration and integration to achieve better financial outcomes and improve supply chain performance. However, various limitations and challenges hinder the utilization of these elements. Effective utilization of supply chain collaboration and integration in food loss and waste management requires members to work collaboratively to manage waste throughout the entire food supply chain network, not just in individual parts.

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Elintarviketoimitusketjuihin kohdistuu merkittäviä paineita tehokkuuden ja kilpailukyvyn suhteen. Tähän vaikuttavat muun muassa erilaiset geopoliittiset, taloudelliset ja ilmastolliset tekijät. Yritykset etsivät tapoja parantaa toimitusketjunjensa tehokkuutta. Yhteistyössä toteutetut ratkaisut hävikin hallinnassa ovat yksi keinoista, jolla elintarviketoimitusketjujen jäsenet voivat vastata näihin tarpeisiin.

Tämän tutkimuksen tavoitteena oli tutkia kuinka toimitusketjuyhteistyötä ja integraatiota voidaan hyödyntää hävikin hallinnassa elintarviketoimitusketjuissa. Tutkimus keskittyi tarkastelemaan erityisesti sitä, miten toimitusketjuyhteistyötä ja -integraatiota voidaan hyödyntää hävikin hallinnassa toimittaja-keskusliike rajapinnassa.

Tutkimus suoritettiin laadullisena monitapaustutkimuksena. Tutkimusaineisto kerättiin puolistrukturoitujen haastattelujen avulla kuudelta Suomessa toimivalta elintarviketeollisuuden yritykseltä. Tutkimusaineiston analysointiin käytettiin sisällönanalyysimenetelmää.

Tutkimuksen empiiriset havainnot osoittavat, että toimitusketjuyhteistyö ja -integraatio ovat hyödyllisiä ja arvokkaita elementtejä hävikin hallinnassa elintarviketoimitusketjuissa. Yritykset ovat kiinnostuneita hyödyntämään näitä elementtejä parempien taloudellisten tulosten ja korkeamman toimitusketju tehokkuuden saavuttamiseksi. Erilaiset rajoitteet ja haasteet muodostavat kuitenkin esteitä näiden hyödyntämisessä. Tehokas toimitusketjuyhteistyön ja integraation hyödyntäminen hävikin hallinnassa edellyttää yrityksiltä kykyä tarkastella hävikkiä läpi koko toimitusketjun eikä ainoastaan ketjun yksittäisissä osissa.

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1 Introduction

Every year a considerable number of limited resources are lost in supply chains as food loss and waste (FLW) (Beretta, Stoessel, Baier & Hellweg 2013, 764). FLW is at the same time a major societal, economic, nutritional and environmental challenge. Economic and population growth has meant that the demand for food has at the global scale increased, which has further put pressure on food supply chains. The challenges related to food waste have been mostly in the past ignored, especially in developed countries due to the abundance of food. (Halloran, Clement, Kornum, Bucatariu & Magid 2014, 294–295) Nowadays, tackling food waste and loss is more critical than ever for the entire global food security. Various geopolitical and climatic events have increased the challenges and risks related to maintaining a sufficient supply of food. (Aminetzah et al. 2022) Simultaneously the challenges of increased competition and constantly changing consumer preferences and behaviours have boosted the need for consumer companies to rethink and redesign their supply-chain operations to make them more resilient and agile (Carr et al. 2021, 1–2). Food retail has in particular been experiencing a drastic business transformation as the industry has been shaped by various trends happening at the same time. Supply chain management is at the heart of managing these trends and in achieving business success. (Småros 2021, 4–5)

In food supply chains (FSCs) food loss and waste occur at every step of the chain (Muth et al. 2019, 1243). Forty-six per cent of all FLW occurs in the downstream part of food supply chains which entails activities such as processing, distribution and consumption. Fifty-four per cent occurs in the upstream part, which includes such activities as production, post-harvest handling and storage. (Food and Agriculture Organization 2013, 12) One of the most significant stages contributing to FLW in food supply chains is the retailer stage (Devin & Richards 2018). Purely from the economic perspective of retailers managing food waste is especially important for the reason that waste generated in the food supply chains diminishes retailers' profits (Småros 2021, 66). Retailers lose a significant amount of money as unsold products are discarded (Tan, Hai, Popp & Oláh 2022, 11). One of the challenges regarding the FLW at the retailer stage is that demand transparency between retailers and suppliers tends to promote the fear of stock-outs which results in FLW in the food supply chain network (Mena, Terry, Williams & Ellram 2014, 153). This is especially true in the

perishable food categories, which usually tend to have shorter self-lives. The poor management of perishable food items in the food supply chains thus has been one of the greatest factors responsible for FLW (Chauhan, Dhir, Akram & Salo 2021, 11).

Efficient management of food supply chains and FLW requires that companies manage their supply chains at the strategic level. A supply chain strategy is therefore essential to manage and reduce waste, achieve competitive advantage and cost-effectively operate supply chain operations (Tan et al. 2022, 5). Strategies which have been associated with improving the supply chain performance are supply chain collaboration (SCC) and supply chain integration (SCI) (Mofokeng & Chinomona 2019). As the current business environment has become increasingly interconnected and networked, it has become clear that companies cannot operate alone, only sharing low levels of information with other members of the supply chains. Taking interactions of other supply chain members into consideration is required in an environment of intense competition and reduced profit margins. Coordination between different supply chain members, such as between retailers and suppliers, is required to achieve effective supply chain management (Neubert, Ouzrout & Bouras 2004, 261; 263). Reduction in the loss in supply chains can often be achieved with improved coordination and collaboration among the members (Rezaei & Liu 2017, 27). SCC and SCI have hence become crucial elements in a successful supply chain (Mathu & Phetla 2018, 2).

Supply chain collaboration and integration have been identified to provide numerous benefits in managing and achieving operational excellence in supply chains. These two have hence become one of the most prominent parts in achieving business success in supply chain management. (Mathu & Phetla 2018, 2) To tackle the FLW food supply chain members should focus and dedicate their resources to accomplishing collaborative and integrated supply chains to reduce the amount of food waste and loss generated in food supply chains (Perry 2019). Although the problem in the food retail sector has been that only a margin of the retailers has been able to comprehensively implement collaborative practices as a key part of their operations (Småros 2021, 63; Elkady, Moizer & Liu 2014, 2). To achieve a truly collaborative way of operating, members of the supply chain must understand and see the value generated from the investment of their time and effort. (Småros 2021, 64).

1.1 Objective of the study and research questions

This study contributes to the future research directions identified by Luo, Olsen, Liu & Zhang (2022, 16). In their study, Luo et al. (2022, 18) note that while the management of food supply chains has been extensively studied and analysed, the research related to food loss and waste reduction in relation to food supply chain cooperation has been relatively scarce. Luo et al. (2022, 18) also emphasize food supply chain integration in FLW reduction as a possible research topic for future studies.

This study also contributes to the research suggestions made by Bergman and Holmqvist (2022, 65–66). Bergman and Holmqvist (2022, 65–66) point out that there is a need for research to study how collaborations function in practice in other food supply chains. They note that more academic, as well as empirical research, is needed to understand how supply chain collaboration could be utilized in an effort to decrease food loss and waste in food supply chains. (Bergman & Holmqvist 2022, 65–66) Also, according to Annosi, Brunetta, Bimbo and Kostoula (2021, 208), previous studies related to food waste have mainly relied on quantitative analysis. According to Annosi et al. (2021, 208), this has meant that the dynamics and tensions related to collaboration between the supply chain members have not received sufficient examination. For these reasons identified from the previous research and literature, it can be concluded that the topic of this study is highly relevant and worth studying from the theoretical perspective as well as from the managerial viewpoint.

The objective of this study is to evaluate the use of supply chain collaboration and integration in managing food waste in food supply chains. The study focuses on investigating how collaboration and integration in the retailer-supplier interface can be utilized to manage and reduce food loss and waste in the food supply chain. The results from this study aim to form a picture of how collaboration and integration could be effectively employed by both retailers and suppliers not only by themselves but in a joint manner in a way which is beneficial for both parties.

One main research question was created to achieve this study's goals. In addition to the main research question, two sub-questions were also formed to support the objectives of this study, add depth to this study, and support the main research question.

The main research question is the following:

Main RQ: How could supply chain collaboration and integration be utilized in the management of food waste generated in food supply chains?

The first sub-question aims to expand the findings of the study by focusing on analysing the benefits and challenges of integration and collaboration in the context of this study. The first sub-question is the following:

SQ1: What are the benefits and challenges of supply chain collaboration and integration in managing food waste within food supply chains?

The second sub-question is related more closely to the incorporation of supply chain collaboration and integration in supply chains. The second sub-question is the following:

SQ2: How can supply chain members incorporate supply chain collaboration and integration into their processes and actions to manage food waste in food supply chains?

1.2 Conceptual framework and the scope of the study

The conceptual framework presented below in Figure 1 shows the main concepts of this study. This study aims to explore the contributions, effects, and relationships of the elements presented in the conceptual framework on the research subject. The conceptual framework will help to analyse and interpret the data collected from the case companies.

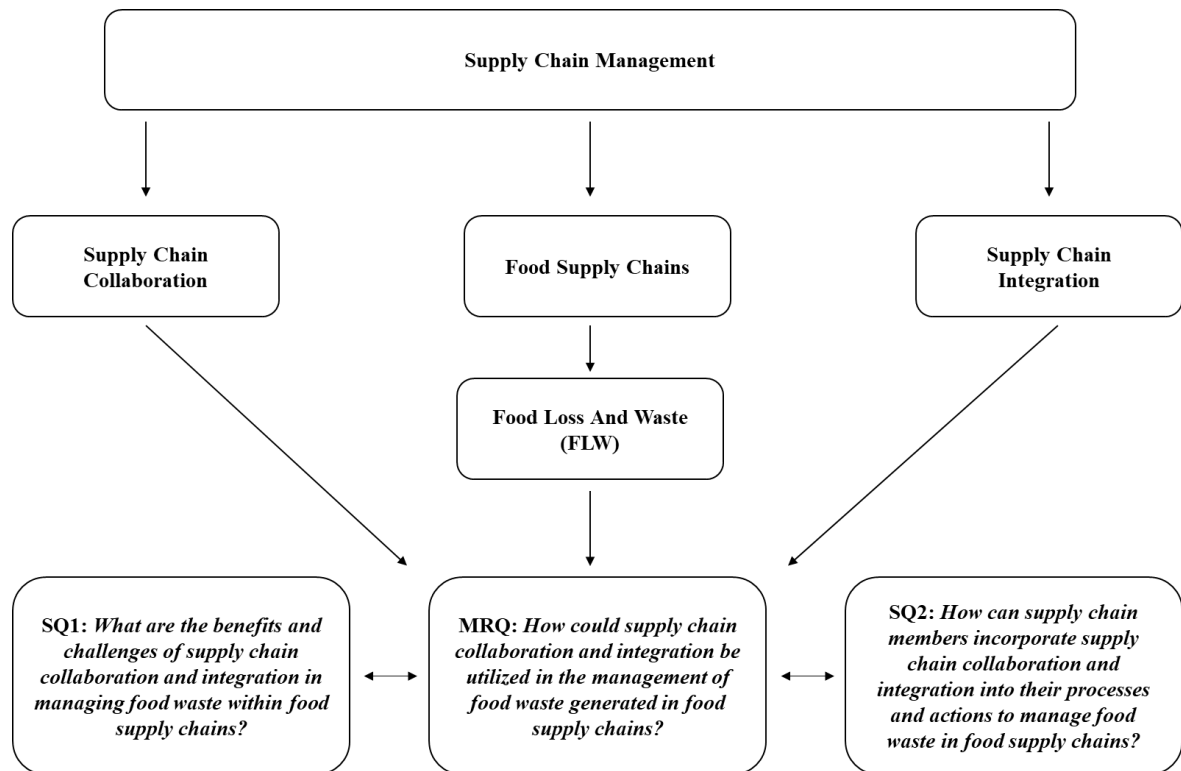


Figure 1. Conceptual framework

There are also some limitations in this study that must be considered. This study examines the subject primarily from the perspective of the retailer-supplier interface. Due to this limitation, some supply chain actors and elements contributing to food loss and waste are exempt from this study. A limitation is also made to focus on developed countries, specifically Finland. This limitation is made since, in developed countries, the level of FLW is higher than in developing countries. In developing countries, FLW occurs primarily due to weak harvesting techniques and poor transport, storage and packing infrastructure. In contrast, in developed countries issue of FLW is related to the problem of exceeding production over demand. (Gustavsson, Cederberg, Sonesson, van Otterdijk & Meybeck 2011, 4–5; 10–11) Another limitation of the study is that only a limited number of product groups are considered in the empirical part of this study. In this study, the focus is to examine perishable fresh produce categories such as dairy and meat -products. Although, the study's findings are not limited to these categories. The goal is to provide solutions, conclusions and proposals that can also be applied to other product groups, such as dry goods. Finally, this study aims not to study consumer behaviour and how it affects food loss and waste in supply chains or its relationship to retailer-supplier collaboration and integration.

1.3 Definitions of the key concepts

Supply chain management (SCM) is defined by Mentzer et al. (2001, 18) as being “*the systemic, strategic coordination of the traditional business functions and the tactics across these business functions within a particular company and across businesses within the supply chain, for the purposes of improving the long-term performance of the individual companies and the supply chain as a whole.*” The Council of Supply Chain Management Professionals (2013, 187) describe supply chain management as being an integrated element that within and across companies integrates supply and demand management processes. Coordination and collaboration with channel partners are important aspects of supply chain management (Council of Supply Chain Management Professionals 2013, 187).

Supply chain collaboration (SCC) is an activity where two or more supply chain members work in cooperation to produce a greater competitive advantage than would be achieved by working in isolation. This competitive advantage is created by the members by sharing information, jointly making decisions, and also sharing the benefits. Supply chain collaboration leads to companies better satisfying customer needs with the increased profitability created in the collaboration by working together. (Simatupang & Sridharan 2002, 19)

Supply chain integration (SCI) can be defined as a comprehensive collaboration in strategic, tactical and operational decision-making in the supply chain network between the members of the SC network (Bagchi, Chun Ha, Skjoett-Larsen & Boege Soerensen 2005, 278). Zhao, Baofeng, Flynn and Yeung (2008, 374) characterize supply chain integration as “*the degree to which an organization strategically collaborates with its SC partners and manages intra- and inter-organization processes to achieve effective and efficient flows of products, services, information, money and decisions, with the objective of providing maximum value to its customers.*”

Food loss and waste (FLW) is defined by Rezaei and Liu (2017, 26) as a “*decrease in the quantity or quality of edible food that is intended for human consumption.*” Although food loss and food waste are often referred to together, they are separate concepts that can be defined apart from each other based on their traits. According to the Food and Agriculture

Organization of the United Nations (2022), food loss refers to the loss occurring in the food supply chain at production, post-harvest, and processing stages. Food loss is the loss of edible food mass in the upstream part of the food supply chain. On the other hand, food waste is the discard of edible food at the downstream parts of the supply chain, such as at the retail and consumer levels. (Food and Agriculture Organization of the United Nations 2022) Food waste is often still fit for human consumption, and the products are removed from the food supply chain and classified as food waste when they have expired or spoiled due to poor stock management. Alternatively, products can be discarded as food waste by choice. (Rezaei & Liu 2017, 26) Food loss mostly occurs in developing countries, while food waste occurs in developed countries (Food and Agriculture Organization of the United Nations 2022).

1.4 Methodology of the study

This study's empirical part has been carried out as a qualitative multiple-case study. According to Creswell (2014, 32), qualitative research is "*an approach for exploring and understanding the meaning individuals or groups ascribe to a social or human problem.*" The purpose of qualitative research is to describe a phenomenon or an event. The qualitative method can be used to understand actions or to form a theoretical interpretation of a phenomenon. (Tuomi & Sarajärvi 2018, 73) A case study method provides an opportunity to gather rich and profound data to be used in qualitative research. The data gathered provides an opportunity to conduct an in-depth analysis of the phenomenon, which has been identified as being useful, especially in the context of the supply relationship, chain and network development. (Kähkönen 2011, 31; 40)

The research data used in the empirical part of this study were compiled from semi-structured interviews. For this study, a total of six individuals from the food retailing industry were interviewed. Five individuals were interviewed from five different supplier companies, and one was from the retailer company. Overall, this study contributes to supply chain management both from the perspective of retailers and suppliers, but the study's findings can also be utilized more broadly in the industry.

1.5 Structure of the thesis

This study consists of six chapters. The introductory chapter presents the subject of the study, along with the study's objectives, research question, and sub-questions in the first subsection. The second subsection provides a conceptual framework and defines the scope of the study. The key concepts of the study are then presented in the third subsection. The fourth subsection briefly introduces the study methodology. The study's theoretical background is presented in the second chapter, starting with research on supply chain collaboration and integration. Next, an overview of food supply chains is established, followed by an examination of the concept of food loss and waste. In the third chapter, the research methodology and research design are introduced in more detail. The fourth chapter presents the study's empirical findings. The fifth chapter, the discussion chapter, analyses the empirical findings in relation to the theory and research questions presented and aims to answer them. Finally, the concluding chapter summarizes the study's findings, presents its limitations, and suggests areas for future research.

2 Theoretical Background

The second chapter introduces the theoretical background of the study. The chapter starts by covering and discussing the topics of supply chain collaboration and integration. The chapter continues by establishing a general overview of food supply chains. After this, the theoretical background of food loss and waste in the context of this study is presented. At the end of this chapter, the issue of food loss and waste in fresh produce items is explored.

2.1 Supply chain collaboration conceptualization

The concept of collaboration has been an extensively studied subject in literature and research (Ramanauskas, Vienažindienė, Rauluškevičienė & Žukovskis 2021, 701). Supply chain collaboration has received a growing amount of interest, especially in the past decade as a way to enhance company and supply chain performance (Panahifar, Byrne, Salam & Heavy 2018, 358). In the context of supply chain management, collaboration can be understood as an effort to attain mutual business goals and objectives with collaborative work efforts and communication between people and systems (Council of Supply Chain Management Professionals 2013, 36). Collaboration can also be understood as a way for companies to build and maintain relationships and communicate with each other (Ramanauskas et al. 2021, 701). Supply chain collaboration can also be seen as a way for companies to leverage their key suppliers' and customers' knowledge and resources to survive in a competitive market environment (Cao, Vonderembse, Zhang & Ragu-Nathan 2010, 6613). As an example, among retailers and manufacturers, a collaboration between the supply chain members is critical for successful supply chain management (Vlachos & Bourlakis 2006, 70).

In scientific literature, various definitions have been established for supply chain collaboration. The concept of supply chain collaboration departs from the broader general concept of collaboration as supply chain collaboration was formed as a concept for supply chain management literature (Bergman & Holmqvist 2022, 24). In its very essence, a collaborative supply chain can be understood as a state where at least two independent

companies plan and jointly execute supply chain operations to achieve greater success than working independently (Simatupang & Sridharan 2002, 19). Based on earlier research and literature, Cao et al. (2010) have formed a comprehensive definition of supply chain collaboration (SCC). Cao et al. (2010, 6616) define supply chain collaboration as “*a long-term partnership process where supply chain partners with common goals work closely together to achieve mutual advantages that are greater than the firms, would achieve individually.*”

Despite the varying definitions for supply chain collaboration, the conceptualization of SCC has typically been two-folded. The supply chain collaboration focus has been divided into process- and relationship integration (Cao & Zhang 2011, 166). The challenge in the SCC definition has been that components of communication and knowledge creation have been in many instances ignored as the focus of supply chain collaboration has mostly been related solely to process integration (Cao et al. 2010, 6628).

2.1.1 Elements and practices of supply chain collaboration

Various recurring and interrelated supply chain collaboration elements have been identified in past SCC research (Bergman & Holmqvist 2022, 22). Based on a review of previous research related to supply chain collaboration, Cao et al. (2010, 6617) argue that there are seven main elements to SCC presented in Figure 2. Cao and Zhang (2011) highlight that these seven interconnecting components form a basis for SCC. This is also supported by Mofokeng & Chinomona (2019, 3) in their research.

Bergman and Holmqvist (2022, 24–25) present a slightly different conceptualization of SCC elements in their study. According to Bergman and Holmqvist (2022, 24–25), the SCC practices consist of resource sharing, joint decision-making, incentive sharing, mutual goals, joint knowledge creation and communication. Although also their conceptualization is built on the research made by Cao and Zhang (2011), who applied the Cao et al. (2010) seven-element approach in their study. For these reasons, in this study, we follow the original conceptualization presented by Cao et al. (2010).

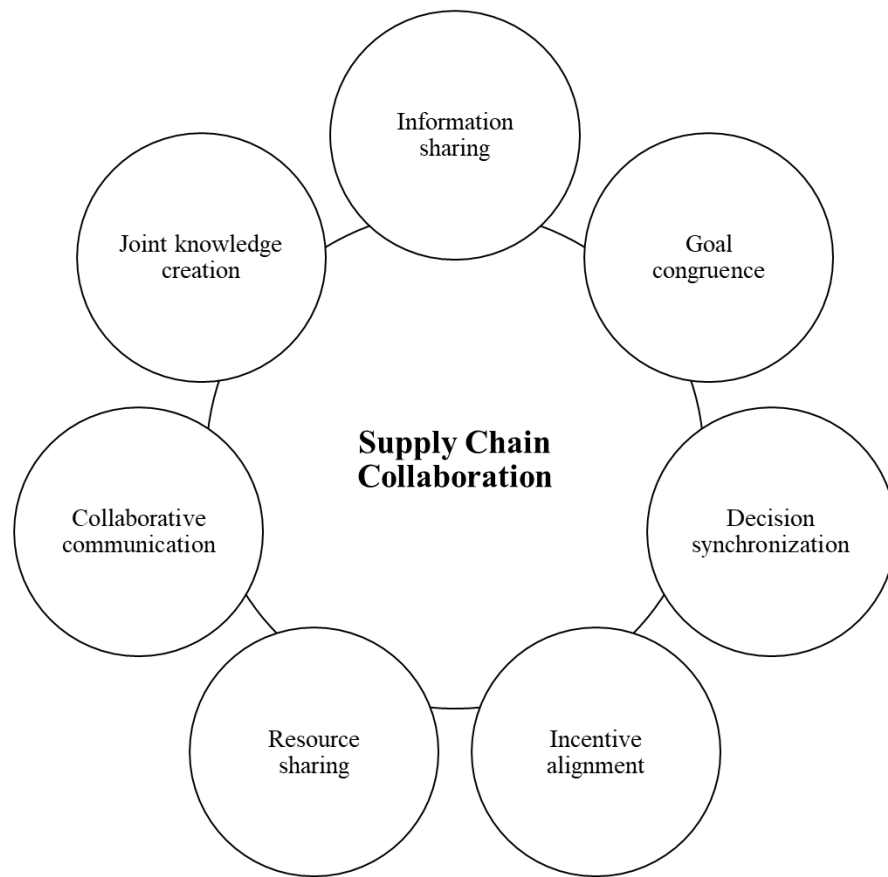


Figure 2. Elements of supply chain collaboration (adapted from Cao et al. 2010)

Information sharing

According to Mohr and Spekman (1994, 139), information sharing refers “to the extent to which critical, often proprietary, information is communicated to one’s partner.” Information sharing is an important factor in achieving effective collaboration in supply chains (Ramanauskas et al. 2021, 701). Bergman and Holmqvist (2022, 25) also highlight information sharing as a key to successful supply chain collaboration. Information sharing promotes adequate visibility across the supply chain functions and members (Simatupang & Sridharan 2002, 24). It is identified as one of the main obligations for collaborative inter-organizational relations (Sheu, Yen & Chae 2006, 27). Sheu et al. (2006, 45) also stress in their research that information sharing is required to improve supply chain collaboration. Collaborations in supply chains are driven by information sharing as well as by trust (Panahifar et al. 2018, 359). For example, information sharing can include exchanging transactional data and sharing customer feedback between members (Wee, Thoo, Sulaiman & Muharam 2016, 2).

Goal congruence

The degree to which supply chain partners believe achieving the supply chain objectives will satisfy their own goals is known as goal congruence (Cao et al. 2010, 6618). In other words, goal congruence is the extent to which businesses believe that achieving a common objective is possible (Jap 1999, 465). According to Jap (1999, 465) goal congruence is “... *the cumulative result of many interactions through which emerge an understanding of each firm's organizational constraints and opportunities.*” Congruence suggests that SCC calls for a certain level of knowledge and agreement among business characteristics, ideas, and practices (Cao et al. 2010, 6618). The definition of roles and responsibilities for each partner, mutual goals, setting objectives, information technology standardization, shared information formalization, as well as joint plan implementation, are all concerned with goal congruence (Min et al. 2005, 244–245). For the long-run survival of a relationship between supply chain members, goal congruence is important because members are driven to sustain collaboration with each other as long as they see that their goals are being met. Strategic collaborations as well as operational collaboration are fostered among supply chain members through goal congruence (Chowdhury, Prajogo & Oke 2015, 5–6).

Decision synchronization

The degree to which chain members can coordinate crucial choices at the planning and execution levels for maximizing supply chain profitability is known as decision synchronization (Simatupang, Wright & Sridharan 2002). Critical supply chain decisions should be coordinated between supply chain members in order to achieve performance increases (Cao et al. 2010, 6619). Decision synchronization is critical in supply chain operations as each member often has varying expertise and decision-making rights. For instance, while a retailer may have the authority to control order quantity, it may not be able to make decisions regarding order delivery. (Simatupang & Sridharan 2005, 264) Decision synchronization among supply chain members helps the members to decide the best possible way to utilize the resources in an effort to accomplish a particular set of objectives. There are several supply chain planning decisions aspects that benefit from decision synchronization amongst the members. These include, for example, operations strategy planning, production planning, production scheduling, procurement, demand management, and distribution management. (Cao et al. 2010, 6619) Decision synchronization covers joint-decision making processes and also decision rights re-allocating (Simatupang & Sridharan

2005, 264). Overall, decision synchronization is utilized to coordinate and align decisions (Cao et al. 2010, 6619). Simatupang and Sridharan (2005, 264) summarize that “... *decision synchronization helps the chain members to carry out productive actions associated with integrated supply chain processes such as replenishment, transportation, and customer service.*”

Incentive alignment

Incentive alignment, according to Simatupang and Sridharan (2004, 488), refers to the process of sharing costs, risks, and benefits among the supply chain members. The costs, risks, and benefits, as well as the generation of incentive schemes, are covered under incentive alignment (Cao et al. 2010, 6619). For the formation of successful supply chain partnerships and the achievement of supply chain collaboration, it is beneficial for all members to equitably share their gains as well as losses (Manthou, Vlachopoulou & Folinas 2004, 241). Incentive alignment motivates supply chain members to aim for effective decisions that strengthen the preferred level of performance. Incentive alignment encourages members to follow the mutual strategic objectives in a constant manner. It also motivates members to make optimal decisions that promote overall supply chain performance. This is achieved in conjunction with decision synchronization. Information sharing is also an essential part of incentive alignment as it conveys to supply chain members that incentives are available, timely, equitable, and performance-based (Simatupang & Sridharan 2005, 264–265). A careful definition of mechanisms that share gains equitably is required in incentive alignment. This means that gains need to be proportional to investments and risks. (Lee & Whang 2001, 5)

Resource Sharing

Resource sharing is described by Cao and Zhang (2011, 166) as “... *the process of leveraging capabilities and assets and inventing in capabilities and assets with supply chain partners.*” Bergman and Holmqvist (2022, 25) mention that both tangible and intangible resources can be shared. Tangible resources are associated with physical components and materials, whereas intangible resources relate to non-physical elements, such as information. Information sharing is thus highly related to resource sharing, and sharing information is recognized as one of the keys to successful supply chain collaboration. (Bergman & Holmqvist 2022, 25)

Collaborative communication

Collaborative communication can be, according to Cao et al. (2010, 6620), understood as “... *the contact and message transmission process among supply chain partners in terms of frequency, direction, mode, and influence strategy.*” Communication is a critical element not only for the development of supply chain collaboration but also for maintaining supply chain collaboration (Chan & Prakash 2012). According to Cao et al. (2010, 6620), communication can be formal or informal. Structured rules and fixed procedures define formal communication compared to informal communication, categorized as non-regularised and spontaneous dialogue (Cao et al. 2010, 6620). Balanced and open two-way communication between members and their operations is a key enabler for close supply chain collaboration (Bergman & Holmqvist 2022, 27). Collaborative communication influences in a positive manner buyer-supplier relationships (Prahinski & Wenton 2004, 59).

Joint knowledge creation

Malhotra, Gosain & El Sawy (2005, 158) define joint knowledge as “... *the ability of business enterprises to increase their knowledge base in conjunction with business partners.*” Joint knowledge creation can be conceptualized as supply chain members' joint action to seek and obtain new and appropriate knowledge to form a better view of the market and, with this knowledge form ability to respond to outside fluctuations and competition in a better way. (Bergman & Holmqvist 2022, 25) Joint knowledge creation consists of two activities: knowledge exploration and exploitation. Knowledge exploration relates to acquiring and searching for new important knowledge, whereas knowledge exploitation focuses on utilizing and applying the gathered knowledge. (Cao et al. 2010, 6620) The knowledge exchange, capture and utilization enable innovation amongst supply chain members and facilitate long-term competitiveness in the network (Harland, Zheng, Johnsen & Lamming 2004, 11). Joint knowledge creation can be identified as one of the main objectives of collaboration (Cao & Zhang 2013, 67).

2.1.2 Motivators and benefits of supply chain collaboration

Companies are motivated to participate in supply chain collaboration as it can provide participating members numerous advantages and benefits that can be highly important in an increasingly challenging business environment (Cao et al. 2010, 6613). Collaboration can

often contribute to improvements in company supply chain efficiency, effectiveness and profitability (Min et al. 2005, 249). In general, SCC allows companies to accomplish better performance (Simatupang & Sridharan 2004, 484).

Companies are motivated to utilize supply chain collaboration for several reasons. Reduced uncertainty, lower transaction costs, generation of core competence, improved competitive advantage, and better utilization of opportunities deriving from knowledge creation and learning are benefits that work as motivators for companies to utilize supply chain collaboration in their operations (Cao et al. 2010, 6613). Collaboration also provides better cost transparency between suppliers and buyers (Gutierrez, Kothari, Mazuera & Schoenherr 2020), better production, higher consumer trust and better capacity utilization (Ramanauskas et al. 2021, 711).

Kumar, Banerjee, Meena and Ganguly (2017, 45) summarize in their research that SCC benefits include a reduction in lead time, inventory, bullwhip effect and an increase in flexibility, end-customer satisfaction as well as in market share and profits. The benefits of supply chain collaboration also expand beyond improved overall profitability and efficiency. For example, better supply chain performance can be achieved as collaboration in supply chains has also been associated with identifying and removing managerial inertia. The removal of managerial inertia helps solve issues related to lower productivity caused by poor supply chain design and supply chain conflict. (Simatupang & Sridharan 2002, 23)

Supply chain collaboration allows supply chain members to take advantage of opportunities to generate profit that could not be achieved individually (Simatupang & Sridharan 2002, 19). One of the most valuable attributes generated with supply chain collaboration is improved responsiveness and agility across the entire supply chain. With supply chain collaboration, inventory levels across the whole supply chain can be lowered, and thus increased efficiency can be achieved. Streamlining distribution and operations costs motivates supply chain management to utilize supply chain collaboration in supply chain operations. (Mathu & Phetla 2018, 2; 5) Demand forecasts and point-of-sale (POS) data sharing from retailers to suppliers are also valuable for achieving efficient supply chain collaboration. Relevant information concerning promotions, events or other changes should

also be shared among members as it gives supply chain members an understanding of retailers' reasoning. (Småros 2021, 63–64)

Bergman and Holmqvist (2022, 29) highlight that motivators of supply chain collaboration can be summarized based on earlier literature into three key traits: increased supply chain knowledge, improved business performance, and solving problems in supply chains. Increased supply chain knowledge helps reduce uncertainties among the supply chain members as they know more about other members' behaviour and decision-making processes. The second motivator, improved business performance, implies that supply chain collaboration improves economic, social and environmental performances. Through supply chain collaboration, the overall performance of the supply chain can be improved, as well as the performance of individual members. Solving problems in supply chains is the third motivator which motivates companies to engage in supply chain collaboration. Supply chain members face supply chain problems in the present and future that affect the members in different supply chain stages. Supply chain collaboration helps solve these issues by incorporating more members into problem-solving. (Bergman & Holmqvist 2022, 28–30)

2.1.3 Challenges of supply chain collaboration

Even though supply chain collaboration provides various benefits, on many occasions' collaborations do not meet their expectations (Cao & Zhang 2013, 1). Supply chain collaboration is rarely without any challenges, as collaboration efforts face both internal and external challenges (Pierre 2020). These challenges may, for instance, be a lack of information, distrust or desire to be independent (Ramanauskas et al. 2020, 711). Companies often face challenges in integrating supplier collaboration into their supply chain and procurement strategies, particularly in the case of supplier collaboration. SCC call for long-term commitment and buy-in from all the members participating in the collaboration. Therefore, SCC should not be viewed solely as the implementation of a framework or procedure. (Gutierrez et al. 2020)

One challenge to achieving supply chain collaboration is the opportunistic behaviour and local perspective that many companies exhibit, which can have negative effects on the overall supply chain profitability (Simatupang & Sridharan 2002, 15). Some collaboration-

breaking characteristics are, for example, retailers' strong focus on delivered price or suppliers' demand to keep tight control of the supply chain (Pierre 2020). Supply chain collaboration can also fail because members do not focus on sharing real-time data but concentrate on forecasting demand on their own. (Pierre 2020) Differences in the mindset of the chain members may prevent collaboration efforts (Benavides, De Eskinazis & Swan 2012). For supply chain collaboration to be successful benefits of SCC should be identified by all supply chain members (Cao et al. 2010, 6628). These cases are examples where supply chain collaboration cannot deliver the desired results (Pierre 2020).

A significant SCC challenge is that members, in many instances, are only accustomed to transactional or adversarial relationships with other supply chain members (Gutierrez et al. 2020). Even though collaboration aims to achieve mutually beneficial goals, companies' participation is often driven by self-interested objectives. In many instances, companies are only interested in participating in collaborative efforts if they see that these efforts can impact their survival. This is problematic as opportunistic behaviour and individual profit maximisation work against overall supply chain performance and profitability, as it is often done at the expense of other supply chain members. This opportunistic behaviour can result in chain members exploiting, for example, outdated policies to maximise their individual benefits. (Simatupang & Sridharan 2002, 15; 17; 19)

Another challenge for SCC is that companies tend to focus on costs instead of value (Gutierrez et al. 2020) which creates a barrier to collaboration. Supply chains that focus on value can simultaneously succeed in various aspects, including flexibility, quality, speed, and cost. In contrast, traditional supply chains often only focus on one aspect, such as cost or speed. (Ketchen & Hult 2007, 573) Change in basic assumptions in the mentality of companies participating in SCC is required if successful collaboration wants to be achieved. This requires a significant dedication of time and management from the companies. Cross-functional involvement from the members typically helps to overcome these barriers and helps the change. (Gutierrez et al. 2020) The benefits that can be formed from supply chain collaboration are often not realised due to differences in business perspective among the supply chain members (Simatupang & Sridharan 2002, 15). That is why all members must understand the benefits of investing time and effort to operate more collaboratively (Småros 2021, 64).

A further challenge for supply chain collaboration is that companies may have trouble quantifying the value generated from the collaboration. Furthermore, on some occasions, companies and managers lack the necessary skills to implement in-depth collaboration programs as they do not have the necessary organisational structure or sufficient amount of skilled personnel to implement them. (Gutierrez et al. 2020) The members participating in the collaboration have differences in their organisational design and culture, making aligning the collaboration efforts challenging. Collaboration can also be problematic if commitment from the senior management for the collaboration is lacking. This can lead to that not enough resources are made available for the collaboration to achieve beneficial results or existing limited resources are being spread over too many initiatives. (Benavides et al. 2012)

Companies' previous experience with bad partnerships can also reflect the collaboration efforts in a way that the members are unwilling to share critical information in the partnership. This leads to suboptimal solutions in SCC (Benavides et al. 2012), and therefore unwillingness to share resources negatively impacts the success of supply chain collaboration (Bergman & Holmqvist 2022, 30). Lack of trust strains the functioning of SCCs, and it may prevent any collaborative practices (Bergman & Holmqvist 2022, 30). Hence issues related to trust are also present in supply chain collaboration (Benavides et al. 2012), similar to supply chain integration. Trust is essential for supply chain collaboration as it promotes transparency, commitment and information sharing among the members (Bergman & Holmqvist 2022, 30).

2.2 Supply chain integration conceptualization

Supply chain integration (SCI) is regarded as one of the key elements in the supply chain management strategy (Cigolini & Rossi 2008) and in achieving excellence in supply chain management (Mofokeng & Chinomona 2019, 5; Saidon, Radzi & Ghani 2015, 1; Hong, Youn & Nahm 2008, 437). Supply chain integration is essential in improving business practices and is crucial for companies to achieve both their economic and environmental goals (Mofokeng & Chinomona 2019, 3). In management practice, supply chain integration has been a fundamental factor in achieving a competitive advantage (Shou, Kang & Park 2022, 1).

The concept of supply chain integration has been understood in previous research and literature in various ways. The concept of integration is at the very core of supply chain management being one of the elements that characterize the concept of supply chain management. (Alfalla-Luque, Medina-Lopez & Dey 2013, 801) The ability of supply chain management to integrate, coordinate, and share information is vital to achieving competitive advantage in supply chains (Neubert et al. 2004, 260). An integrated supply chain can be described as a strategic collaboration between a company and its supply chain members to manage collaboration in intra- and inter-organizational processes (Flynn, Huo & Zhao 2010, 58). National Research Council (2000, 27) define an integrated supply chain “*as an association of customers and suppliers who, using management techniques, work together to optimize their collective performance in the creation, distribution, and support of an end product.*”

One of the challenges with the concept of integration is that it is used interchangeably with cooperation and collaboration, which makes defining and using the term a challenge (Adams et al. 2014, 302). Compared to cooperation and collaboration, integration can be distinguished as an activity which focuses on distributing, aligning and utilizing internal and external resources more efficiently by restructuring company processes (Chen, Daugherty & Roath 2009, 66–67). According to Chen et al. (2009, 75), this is achieved by simplifying activities and with process and informational connectivity. Chen et al. (2009, 64) note that connectivity and simplification can be held as the most basic but also as the two key elements for integration.

In today’s networked and interconnected business environment, the role of integration is more crucial than ever. Technological advantages and their adoption to supply chain management research in past years has meant that there is no longer a need to limit supply chain integration to either upstream or downstream integration (Shou et al. 2022, 10). Lines between planning and execution in supply chains have become blurred as they have become constantly more dynamic and real-time (McDonnell, Sweeney & Kenny 2004, 15).

2.2.1 Dimensions and elements of supply chain integration

Supply chain integration can be divided into three dimensions: internal-, supplier- and customer integration (Shou, Li, Park & Kang 2017, 297). Supply chain integration is often divided into internal and external integration, as supplier and customer integration are frequently referred to as external integration (Flynn et al. 2010, 59). This study analyses supply chain integration from internal and external dimensions.

Internal integration focuses on the company's internal departments and functions. Internal integration drives internal departments and functions to work as an integrated process to achieve the best possible supply chain value. (Flynn et al. 2010, 59) With internal integration, a company can attain better use and coordination of their internal resources (Wong, Wong & Boon-itt 2013, 568). Internal integration enables companies to meet customer requirements more efficiently, which means better performance overall. Internal integration also generates collaboration and breaks functional barriers. (Flynn et al. 2010, 60)

External integration focuses on the value of relationships between the supply chain members. External integration focuses on building interactive and close relationships with the members to achieve maximum value in the supply chain (Flynn et al. 2010, 59). Supply chain integration has evolved from solely focusing on internal supply chain integration to focus more on external supply chain integration. This transition has occurred as companies have realized that only a limited amount of performance improvement can be achieved by focusing only on internal supply chain integration. Performance improvements with SCI often are hard to achieve without the involvement of suppliers and customers. (Stevens & Johnson 2016, 30–31)

Stevens (1989) presents in his study a 4-stage integration framework. According to Stevens (1989, 6–7) these four stages are the baseline stage, stage of functional integration, stage of internal integration and stage of external integration. At the first stage in the so-called baseline stage, supply chain activities and challenges are referred to as separate independent departments inside the company. Supply chain operations are not integrated, and control systems and procedures are independent and incompatible with one another. The stage is

characterized by strong organisational boundaries and by a high level of fragmentation. Each company function only works to achieve its individual goals and objectives. (Stevens 1989, 38) It is important to note that in today's highly competitive environment, where competition among companies and their supply chains is more critical than ever (National Research Council, 2002), the baseline stage approach to integration is no longer recommended.

Stage two of functional integration is characterised by an emphasis on cost reduction in the supply chain. At this stage, company departments are working together to reduce costs. Performance improvements are not considered in the functional integration stage. Internal trade-offs are made, and customer service is still primarily reactive. An entire company level of integration characterises the third stage of internal integration. The internal integration stage focuses on efficiency over effectiveness, but at this stage, the focus remains on tactical rather than strategic issues. Departments at this stage are connected to each other via resource planning systems, and information technology is utilized extensively to attain data interchange across departments. (Stevens 1989, 3–8) When it comes to digital technologies and digitalisation, they have been recognised as a way to improve internal efficiency in supply chains and achieve better company performance (Liu & Chiu 2021, 374; 382). According to Liu & Chiu (2021, 384), digital technologies and investments towards them should be directed especially towards internal integration processes. Furthermore, some critics have argued that supply chain integration research has focused solely on the integration between the customer and supplier and that the importance of internal integration has been set aside (Flynn et al. 2010, 58).

A complete supply chain integration is achieved at the final stage of external integration. Integration scope expands over company boundaries to take into consideration other supply chain members like suppliers and customers. The focus on the integration stage moves from a product-orientated approach to a customer-orientated approach. At the stage of external integration, the focus shifts from the adversarial attitude of conflict to cooperation between members. All supply chain members work jointly to achieve mutual goals and provide final customers high-level customer service and added value. (Stevens 1989, 3–8) It should be remembered that even though internal and external integration has different roles in supply chain integration, both are highly important for successful SCI. The supply chain's best

possible value can only be achieved if both perspectives are considered. (Flynn et al. 2010, 59)

2.2.2 Motivators and benefits of supply chain integration

Several factors drive integration in supply chains, such as increased cost competitiveness, shorter product life cycles, faster product development cycles, globalisation, customisation of product offerings and higher overall quality (National Research Council 2000, 28). Integration in supply chains is typically taken on when operations in the buying organisation or the supply network require improvement (Vachon & Klassen 2006, 800). Reduced redundancy, better management of complexity and improved efficiency of interfirm relationships can be achieved by embracing integration amongst supply chain members (Chen et al. 2009, 75). Supply chain integration is a way to achieve smoother and more reliable material, and information flows through the supply chain (Shou et al. 2022, 23). Management of multi-echelon inventory flows is possible with an integrated supply chain. This helps to minimise the waste further and obtain a high level of automation in the supply chain. Seamless optimisation of inventory flows in the supply chain can be achieved when in the same planning systems demand forecast-, stock-, delivery schedules-, lead times and batch sizes data are made available for all supply chain stages through supply chain integration. (Småros 2021, 46)

Organisational boundaries between functions and barriers between organisations can also be broken with supply chain integration (Alfalla-Luque et al. 2013, 802). Increased competitiveness and profitability are often identified as the key potential benefits of supply chain integration. The ability to react faster to fluctuating market demands, reduced waste of resources, a better synergy between supply chain members, and lower transaction costs are of the few benefits that can be achieved with supply chain integration. (National Research Council 2000, 33) Other benefits of SCI are, for example, supplier collaboration as well as supply chain transparency that supports capacity planning (Småros 2021, 44).

2.2.3 Challenges of supply chain integration

Companies in supply chain integration are bound by trust, shared objectives, and contracts, but only on a voluntary basis (National Research Council 2000, 27). Thus, trust is often the most challenging aspect of supply chain management and integration (Katunzi 2011, 108). Challenges in trust typically relate to the fact that participants are unwilling to share their sensitive data with external members. Another challenge related to trust is that members of the supply chain might not see the value in one another's data that is being shared between the members. For supply chain integration to be effective, the terms and incentives of data sharing must be agreed upon, though it can also make the integration between members challenging. (Ebel et al. 2021) One of the key drivers in achieving supply chain integration success is that quality information is shared extensively between the supply chain members. Establishing partnerships with the chain members is thus incredibly important for successful supply chain integration, as only then is information shared sufficiently between the members. (Hong et al. 2008, 450)

Another critical aspect to remember when implementing integration into supply chain operations is that beneficial results in supply chain performance can only be achieved if the integration actions are carried out in the organisation's context (Mofokeng & Chinomona 2019, 4–5). Furthermore, the implementation and design of supply chain integration cannot be only short-term aimed or done to achieve monetary goals. SCI implementation and design require long-term and strategic planning in an effort to achieve beneficial results. (Huo, Qi, Wang & Zhao 2014, 378). In the implementation of supply chain integration, it should be taken into consideration that in order to take advantage of businesses opportunities deriving from the integration every member participating in the integration need to direct their management skills, processes and technologies into significant functions and capabilities of the supply chain in a disciplined manner. Only then can benefits such as increased profits and lowered risks be achieved. (National Research Council 2000, 27)

2.3 Overview of food supply chains

A food supply chain (FSC) can be defined as *“a set of interdependent companies that work closely together to manage the flow of goods and services along the value-added chain of*

agricultural and food products, in order to realize superior customer value at the lowest possible costs.” (Folkerts and Koehorst 1997, 11) FSCs are complex chains of different elements (Yadav, Singh, Gunasekaran, Raut & Narkhede 2022, 686). Complex issues such as a large number of stakeholders, the perishable nature of products and inter-sectoral influence make food supply chains different from other supply chains (Mithun Ali et al. 2019). Food supply chains cover various sectors and involve several different stakeholders at various stages. Agriculture, industrial processes, retail and waste are examples of sectors related to food supply chains (Porter, Reay, Higgings & Bomberg 2016, 722). Each actor in a FSC has distinctive impact on how a product is produced and distributed for the final customer. Each actor has a business model, which they follow and aim to implement in the best possible way. At each stage of the FSC, the supply chain actors add their value(s) to the final product. (Nosratabadi, Mosavi & Lakner 2020, 2) Therefore, one of the goals of the FSC is to add value throughout the chain (Thomé, Cappellesso, Ramos & de Lima Duarte 2021, 5). In summary, the primary goal of the FSC is to refine a set of raw materials into a final product that satisfies the final customer's needs at the end of the supply chain (Nosratabadi et al. 2020, 2).

According to Tan et al. (2022, 2), all food activities, from farm to table, can be seen to be a part of the food supply chain. Usually, the FSC consists of five distinct factors which are presented in Figure 3. Each factor has a unique role in the FSC and how value is added to a final product. The first stage of the food supply chain involves initial production and harvesting, which are performed by the producers. In the second stage of the FSC, final products are manufactured and packaged by the processors. The third stage is managed by distributors, whose aim is to supply the final products manufactured by the processors to the retailers. At the end of the FSC, in the fourth and fifth stages, retailers sell the final product to consumers. (Nosratabadi et al. 2020, 2) The first three stages are considered upstream stages of the food supply chain. The distribution and the consumer stages are considered downstream stages of the FSC (Porter et al. 2016, 723).

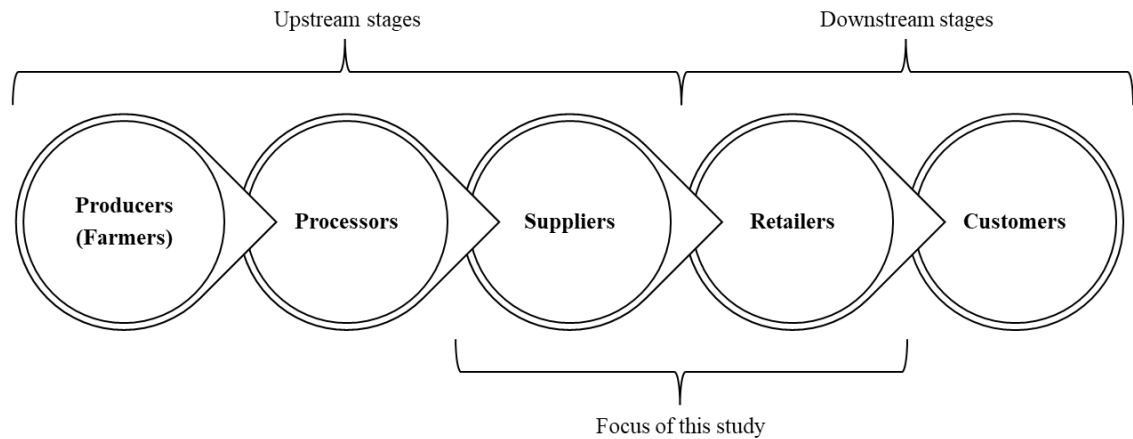


Figure 3. Food supply chain factors (adapted from Nosratabadi et al. 2020; Porter et al. 2016)

Although, it should be noted that the role of distributors has been weakened since nowadays most manufacturers distribute their products directly to retailers. This has meant that role of distributors as a middleman in the food supply chain has been replaced by the manufacturers' direct distributing. (Bourlakis & Weightman 2004, 8) FSCs are also affected by various other stakeholders, such as government agencies, non-government organizations (NGOs) and shareholders (Yadav et al. 2022, 685).

The food supply chain consists of three different flows. These are the produce flow, the finance flow and the information flow. Produce flow moves from the upstream stages of the supply chain to the downstream stages. The first stage of the produce flow covers the producers, and the last stage covers customers at the end of the chain. Financial flow moves in the opposite direction in the food supply chain. Finance flow moves from the downstream to the upstream stages, the first stage being end consumers and the final being producers. The flow of information in the food supply chain is more complicated than produce and finance flows. Each actor in the food supply generates information. End customers generate demand information which flows to the retailers. Aggregated demand information is moved from retailers to distributors. The distributor stage is held as a junction of supply and demand in the food supply chain. The flow of produce availability information goes from producers to processors. From processors, aggregated produce availability information flows to the distributors. (Yadav et al. 2022, 685–686) Figure 4 below presents an illustration of the flows within the food supply chain.

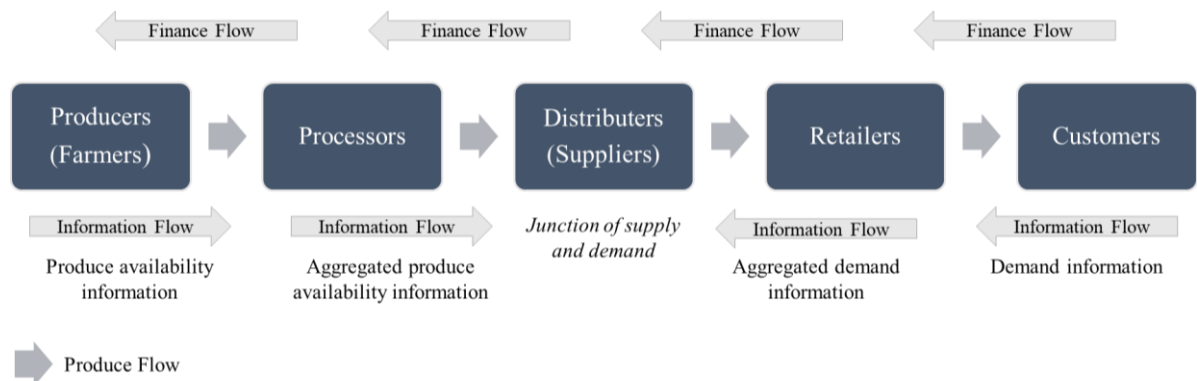


Figure 4. Flows in the food supply chain (modified Yadav et al. 2022, 686)

In the past, FSC phases were seen more as individual elements, each motivated only by profit maximization (Folkerts & Koehorst 1997, 12). Food supply chains were shorter and more independent (Bourlakis & Weightman 2004, 1). Nowadays, individual way of operating has mostly been abandoned as the food supply chain has become an interconnected system affected by various processes performed in complex relationships by different organizations (Folkerts & Koehorst 1997, 12). These days FSCs are considered coherent and unified relationships between the chain members (Bourlakis & Weightman 2004, 1). This is because, in the agribusiness and food industries, it is constantly more challenging for individual companies to succeed independently (Folkerts & Koehorst 1997, 11–12). FSCs have shifted from the individual perspective to a more holistic approach focusing on efficiency and effectiveness (Zhong, Xu & Wang 2017, 2086).

2.4 Food loss and waste

Food loss and waste (FLW) is a critical global concern that has significant environmental, social, and economic implications. The loss and wastage of food have far-reaching impacts. (Surucu-Balci & Tuna 2019, 278) From the environmental point of view, FLW contributes greatly to climate change. From the social perspective, food loss and waste are issues as people suffer from malnutrition and hunger while food produced for human consumption ends up being wasted. From an economic point of view, food loss and waste cause various inefficiencies. (Närvänen, Mesiranta, Mattila & Heikkinen 2019, 2) Economic, social and

environmental concerns have increased the need to manage and reduce FLW. This need has increased interest in optimizing food supply chains and further research issues related to food loss and waste, especially from the operational perspective. (Luo et al. 2022, 2) Despite this, the amount of FLW remains high even though efforts have been taken to reduce and minimise food loss and waste (Chauhan et al. 2021, 1). Every year worldwide approximately 1.6 billion tons of food is wasted which converts in monetary terms to 1.2\$ trillion (Surucu-Balci & Tuna 2019, 278).

The Natural Resources Institute of Finland (2021) study found that in Finnish food supply chains every year approximately 640 million kilos of food go to waste. From this amount, about 360 million kilos are categorized as waste that would have been suitable for human consumption (Luonnonvarakeskus 2021). When examining the relative food waste by retailers it can be noted that retailers operating in the Finnish grocery market have relatively similar levels of food waste. The relative percentage of food waste generated by Finnish grocery retailer S-Group was back in 2021 at 1,47 percent (Nikander 2022). The second largest Finnish retailer K-Group reported that their relative food waste percentage in 2021 was 1,23 percent. The third largest retailer LIDL reported that their food waste percentage in 2021 was 1,30 percent. (Nikander 2022) These values should be compared with care as there seem to be differences in how retailers calculate and evaluate the amount of waste generated. One of the challenges of FLW is that the topic lacks unified and exact measurements for quantifying the waste generated as unified metrics and measurement techniques have not been defined (Borens, Gatzner, Magnin & Timelin, 2022).

Previous studies of FLW have suffered from the lack of standardized definitions in FLW terminology (Garcia-Herrero et al. 2018, 26; Chauhan et al. 2021, 3), and for this reason, terms are often used interchangeably in the research (HLPE 2014, 21–22; Chauhan et al. 2021, 3). FLW can be labelled as an unstructured problem due to this lack of shared definitions (Närvänen et al. 2019, 2). According to HLPE (2014, 22), food loss and waste *“refers to a decrease, at all stages of the food chain from harvest to consumption in mass, of food that was originally intended for human consumption, regardless of the cause.”* The traditional approach to FLW has focused mainly on the food discarded or not used (Irani et al. 2018, 368), but in reality FLW comprises various dimensions making it a fairly complex subject (Chauhan et al. 2021, 3). From the supply chain perspective terms can be

distinguished based on where the waste occurs in the supply chain (Surucu-Balci & Tuna 2019, 278; Luo et al. 2022, 3). Food waste typically occurs at the retail and customer stages whereas food loss occurs in the production, post-harvest and processing stages of the food supply chain (Rezaei & Liu 2017, 26). Loss in quantity and quality at the upstream part of the food supply chain can be understood as food loss. Food waste occurs at the downstream stages of the chain and includes consumer- and retail consumption. Most of the research related to FLW has focused on the consumption stage and the downstream part of food supply chains. (Luo et al. 2022, 3–4; 8) Also, a vast number of solutions to reduce food loss and waste have focused on changing individuals' attitudes and behaviours (Närvänen et al. 2019, 4).

Waste and loss of food occur at every stage of the food supply chain (Göbel, Langen, Blumenthal, Tetscheid & Ritter 2015, 1429; Chauhan et al. 2021, 1). From the perspective of food supply chains decreasing FLW is important as it improves the supply chain efficiency (Parfitt, Barthel & Macnaughton 2010). Also, management and mitigation of FLW is incredibly important as they have a critical role in ensuring food security for the future (Surucu-Balci & Tuna 2019, 278). Additionally, FLW can be connected to increasing food prices as loss and waste of food decrease the amount of food in the market. This is a concern as the increase in food prices has a negative impact, particularly on low-income consumers' ability to access food. (Rezaei & Liu 2017, 26) It is important to note that the concept of food loss and waste includes not only the physical products that are discarded but also the valuable resources that are wasted as a result of FLW. For example, resources such as water, energy and labour are also lost when food turns into waste. (Surucu-Balci & Tuna 2019, 278)

FLW issues at the upstream part of food supply chains are typically impacted by problems related to infrastructure and technological challenges. Infrastructure problems can be, for example, poor network conditions, inadequate technologies or poor infrastructure in general. On the downstream part of the food supply chain issues of FLW mainly comprise incorrect human behaviour and decisions. (Surucu-Balci & Tuna 2019, 279) It is also good to understand that the generation of FLW varies between countries and regions. In developing countries, the majority of the FLW occurs at the harvest, sorting, grading and processing

stages of the food supply chain. On the other hand, most waste in developed countries mainly occurs at the consumption stages. (Rezaei & Liu 2017, 26)

Reasons leading to FLW in food supply chains can be categorized into three categories. These categories are micro-level causes, meso-level causes and macro-level causes. Micro-level causes relate to issues of packaging and transportation. Meso-level causes are issues related to infrastructure and FSC practices, while macro-level causes relate to purchasing options. In previous research, it has been shown that cause of FLW has been typically divided so that either an FLW is seen as a result of a single particular issue in the food supply chains or, on the other hand, it is seen as a result of multiple issues which lead to FLW in the food supply chain. Particular factors contributing to the generation of FLW in food supply chains can be, for example, stakeholder attitude, buyer-supplier agreements, supply chain interruptions, improper packaging or stringent quality standards. Studies focusing on multiple factors mention that significant sources of FLW are, for example, lack of skilled labour, quality errors, limited innovation, lack of regulations and low technical expertise. (Chauhan et al. 2021, 2; 5–6) According to Suruzu-Balci and Tuna (2022), FLW drivers can also be divided into production stage drivers, processing stage drivers, and distribution and retail stage drivers. The production stage contains logistics-, production-, environment- and human-related drivers. The processing stage also comprises logistics-, environmental-, and human-related drivers, but it also considers processing- and standards-related drivers. The distribution and retail stage consists of logistics-, standard- and human-related drivers. Also, marketing-, and management-related drivers are considered at this stage. (Suruzu-Balci & Tuna 2022, 283–287)

Wastage of food in food supply chains is a multifaceted problem (Cakar, Aydin, Varank & Ozcan 2020, 1). Causes that lead to FLW in food supply chains expand over various chain levels and differ from stage to stage. Waste generation and production also varies between different products and product groups (Göbel et al. 2015, 1441). Närvänen et al. (2019, 5) mention that “... *food waste occurs at the intersection of several influences across the food system.*” This signifies that the attention of various stakeholders involved in FSCs is required in order to manage food loss and waste (Chauhan et al. 2021, 2). Mitigation of FLW cannot solely focus on a specific stage or aspect. Solutions to reduce FLW should be planned and implemented in conjunction with other stages and actors, not in isolation. (Strotmann et al.

2017, 2) An effective action plan is needed for retailers and manufacturers to mitigate their FLW. In cooperation with each other, all FSC members should establish a baseline and set targets for FLW reduction. Systematic development and implementation of initiatives follow the baseline and target establishment. Finally, companies need to put the enablers that nurture lasting change in FSCs in place. FLW reduction requires a paradigm shift from the FSC companies. Companies need to understand and see that FLW contributes to inefficiencies in their operations and that across various departments, opportunities are being missed. (Borens et al. 2022) Good relations between the food supply chain members should be stressed, as bad relations are one reason that leads to FLW (Surucu-Balci & Tuna 2019, 289).

Food loss and waste reduction in food supply chains should encounter inefficiencies within FSCs as a whole. The European Commission also proposes this approach to FLW reduction (Hartikainen, Mogensen, Svanes & Franke 2018, 503). Courses of action need to be taken by each food supply chain member in cooperation with other members in other stages of the food supply chain to decrease the amount of FLW generated. A decrease in FLW can be only achieved with shared responsibility among the supply chain members and with the connection of activities which have an impact on other members in the food supply chain both on a regional and a global scale. Recognition of interlinkages between the stages of the food supply chain is at the core of forming solutions to manage better and reduce food loss and waste. This means that the product quality further along the food supply chain depends on how each actor performs. (Göbel et al. 2015, 1441) Retailers along with food manufacturers can, in general, utilize four different controls to control FLW. These are minimising loss during production and processing, minimising loss during transit, selling more of what is produced and structurally preventing loss. Each of these controls consists of various actions and implementation of these controls calls for a substantial dedication of resources from the FSC companies. (Borens et al. 2022)

Various plans and strategies have been identified in research and literature that could be utilized in reducing food loss and waste in FSCs. In FLW reduction, some strategies and attributes, such as using digital technologies to make supply chains more traceable, have been identified as best-in-class practices in FLW mitigation (Borens et al. 2022). To map out the FLW 'hot spots' in the supply chain an active and constant engagement between food

supply chain members particularly between retailers and suppliers is required. Forming an integrated database and providing access to this database can also be beneficial in FLW management. Other enablers for reducing FLW in food supply chains are collaborating with suppliers to match supply and demand better, overhauling existing procurement practices, finding new creative ways to turn loss and waste into value and most importantly enabling true and lasting change. (Borens et al. 2022)

2.5 Food loss and waste in fresh produce supply chains

Fresh produce items pose unique challenges in food supply chains. According to Riipi et al. (2021, 26), a study on FLW by product categories in 2019 found that bread and bakery products accounted for the highest quantity of FLW. The second most waste was produced from fresh vegetables, roots and fruits and third most from meat and fish products. The fourth biggest contributor to waste was milk and milk products, cheeses, fats and eggs. (Riipi et al. 2021, 26). Findings by Riipi et al. (2021, 26) are also illustrated in Figure 5. On the other hand, Horoś and Ruppenthal (2021, 13) identified in their study that fruits and vegetables produce the highest level of waste, followed by milk and dairy commodities which were followed by meat and sausages. It can be understood that waste generation is context, region and country dependent. Therefore, variation regarding the categorisation differs between studies.

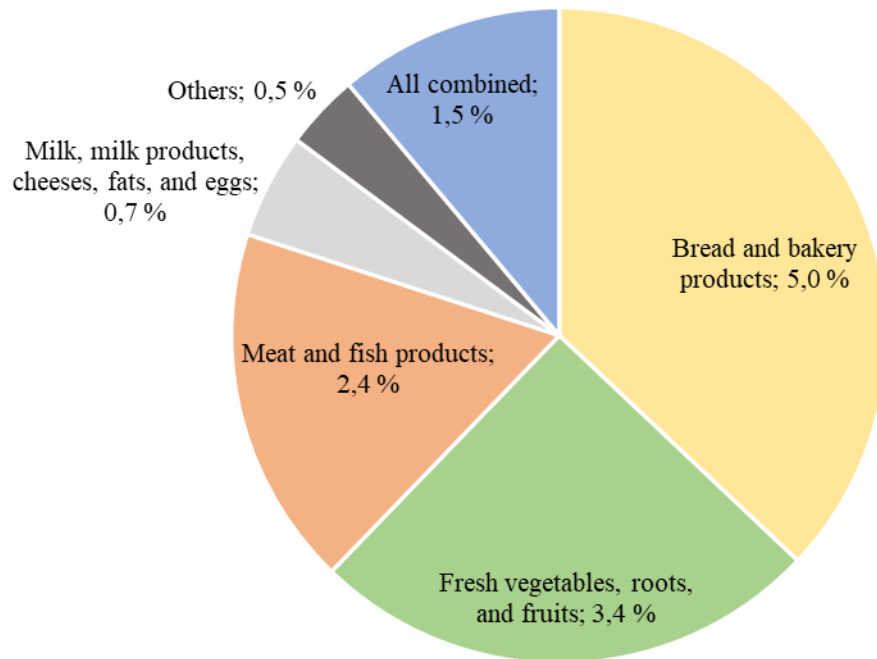


Figure 5. The share of FLW (%) in the total sales volume of Finnish retail trade and wholesale trade in 2019, broken down by product groups (modified from Riipi et al. 2021, 26)

A significant element contributing to FLW in food supply chains is poor management of perishable food products (Chauhan et al. 2021, 11). Products with long shelf lives require operational efficiency from the food supply chain. Minimising costs and maximising availability is important for these kinds of products. Perishable products on the other hand require a responsive supply chain that can yield high availability while minimising waste and markdowns. (Småros 2021, 5–6) The accurate forecasting of demand for fresh products is highly critical due to the perishable nature of these goods (Thakkar & Ravalji 2019, 2). For perishable products, accurate and efficient demand and replenishment are thus extremely important. Compared to other products supply chain planning for perishable items should be available to capture even the most minor changes in demand and adapt supply chain actions accordingly. This differs from products with a long shelf life as they do not require immediate synchronisation with the demand. (Småros 2021, 5–6)

Due to their short shelf-life and inaccurate forecasting, it is almost evitable that waste is formed from perishable products (Mena et al. 2014, 151). Fluctuations in demand can cause a vast amount of waste of perishable products. For example, due to COVID-19 in the first

half of 2020 several tons of perishable products were discarded due to a decrease in demand. (UNECE 2020)

For fresh perishable products, it is essential that the products can be delivered to customers as quickly as possible to maximise products' shelf lives (Surucu-Balci & Tuna 2019, 283). This is important as reduced shelf lives of perishable products will result in customers not buying the products, which means that products likely turn into waste (Göbel et al. 2015, 1434). Therefore, with some fresh products due to their short shelf life aiming for 100%, on-shelf availability is not feasible. Supply chain management needs to find a balance between waste and stock level. Optimisation of supply chains and replenishment is needed to find this optimum point of stock level and waste. Typically, retailers' challenges with perishable products are either that too large orders are being made that cause waste, too much shelf space at the stores is allocated for the product or due to systematic patterns of poor performance on some specific weekdays which result in high waste. (Småros 2021, 27–28)

3 Methodology and Research Design

The objective of this chapter is to provide a detailed description of the chosen research methodology and design. This chapter introduces the research method used in this study and explains why the method in question has been selected. The chapter begins by discussing the chosen methodology and approach to this study. Following the discussion, case selection, research data collection and data analysis will be described. The study's reliability and validity are also reviewed at the end of this chapter.

3.1 Research methodology and process

The first methodological choice in a research process is to choose the research method. Typically, either a quantitative or qualitative research method is chosen for the research. Research can also follow a mixed methods research design which can be interpreted as a combination of quantitative and qualitative methods. (Saunders, Lewis & Thornhill 2016, 164) Quite often, qualitative research is defined in comparison to quantitative research (Eriksson & Kovalainen 2008). One way to distinguish qualitative research method from quantitative is that while quantitative research concentrates on numeric data, qualitative research focuses on non-numerical data, emphasizing words rather than numbers (Bryman & Bell 2011, 386; Creswell 2014, 32; Saunders et al. 2016, 165). Historically, quantitative research has been the dominant method used in business research (Eriksson & Kovalainen 2008). Nevertheless, both methods should be used in academic research to ensure the diversity and profundity of the academic research findings (Kähkönen 2011, 31).

Creswell (2014, 32) defines qualitative research as *“an approach for exploring and understanding the meaning individuals or groups ascribe to a social or human problem.”* Qualitative research is interpretive since it aims to understand the subjective and socially constructed meanings associated with the phenomenon being investigated (Saunders et al. 2016, 168). In qualitative research, the interest is often in the experiences, thoughts and feelings of the people who are the research subject (Puusa, Juuti & Aaltio 2020). Qualitative research is conducted in the natural environment of the research subject to gather detailed

data that is then used to form themes and make interpretations about the phenomenon (Creswell 2014, 32; Saunders et al. 2016, 168). A further distinction that can be made is that this study is exploratory since it aims to gain a deeper understanding of a given issue, problem or phenomenon (Saunders et al. 2016, 174–175).

Illustrated in Figure 6 are the typical six main stages of qualitative research. The first stage of qualitative research is forming general research questions (Bryman & Bell 2011, 389). The research begins with the researcher's idea to form a study on a subject that the researcher sees as a possible research topic (Puusa et al. 2020). This stage is followed by selecting the relevant site(s) and subjects. The third stage of the research is the collection of relevant data. The fourth stage of qualitative research is data interpretation, followed by conceptual and theoretical work. This fifth stage may also include a tighter specification of the research question(s) and further data collection. (Bryman & Bell 2011, 389–392) Due to its flexible nature, the qualitative research method allows the researcher to return to the choices made and alter them if necessary (Puusa et al. 2020). The last stage of qualitative research consists of writing up findings and conclusions (Bryman & Bell 2011, 389–392).

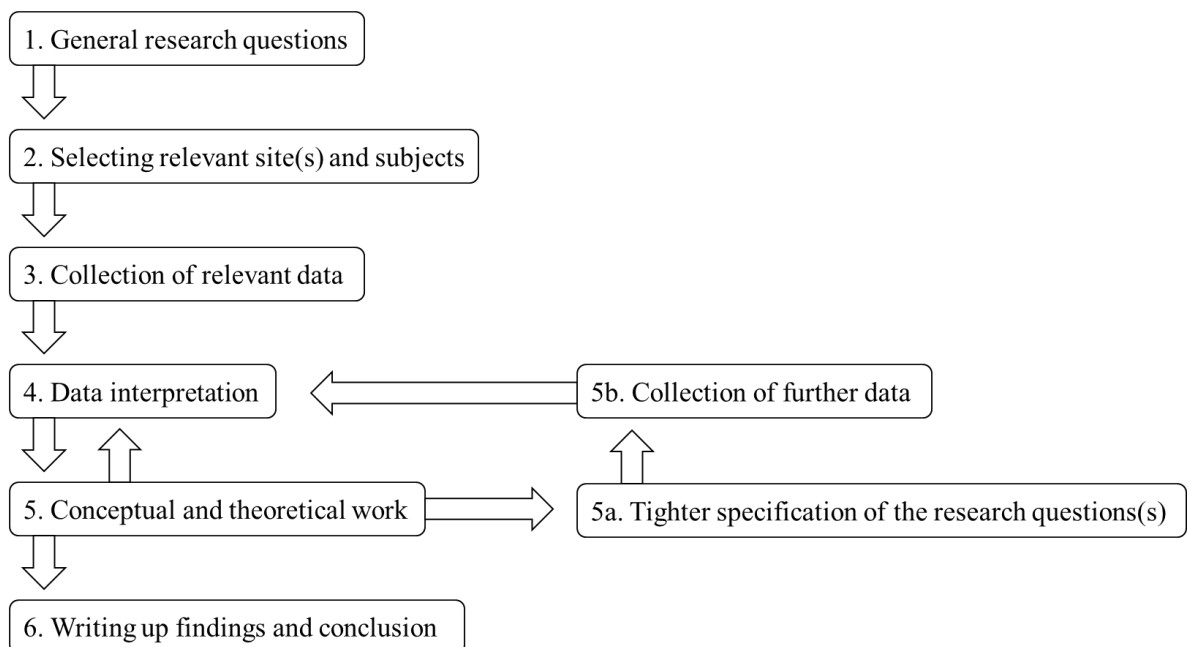


Figure 6. Steps in qualitative research (adapted from Bryman & Bell 2011, 389–392)

This study aims to interpret and identify how retailers and suppliers can utilize supply chain collaboration and integration to reduce food waste and loss in food supply chains. Qualitative

methods such as interviews can provide valuable insight into how the supply chain members perceive and experience food waste management. Qualitative methods can also provide detailed information about how people think, feel and behave with regard to food waste management. Qualitative methods are particularly useful in exploring the social and cultural aspects of food waste management and the complex relationships between stakeholders in the food supply chain. Qualitative research methods can also provide valuable insight into the barriers and facilitators to effective food waste management with supply chain collaboration and integration. Therefore, given the nature of this study, a qualitative, exploratory approach is considered appropriate. This study's aim, objective, and research questions also support using a qualitative research approach.

3.2 Case study approach

This study is conducted as a case study. In its essence, a case study is a design of inquiry (Creswell 2014, 43). Yin (2018, 349) defines a case study as “*a social science research method, generally used to investigate a contemporary phenomenon in depth and its real-world context.*” Saunders et al. (2016, 711) add that a case study involves a variety of sources of evidence used to investigate the phenomenon in question. In a case study, the term “case” can refer to various subjects, including but not limited to individuals, groups, organizations, associations, change processes, or events. To gain a comprehensive understanding of the research subject in a case study, it is crucial to comprehend the dynamics of the topic and the interactions between the research subject and the study's context (Saunders et al. 2016, 184). Based on previous literature, Kähkönen (2011, 32) proposes that the case study research process consists of eight distinctive phases. The research process for case studies proposed by Kähkönen (2011, 32) is illustrated in Figure 7.

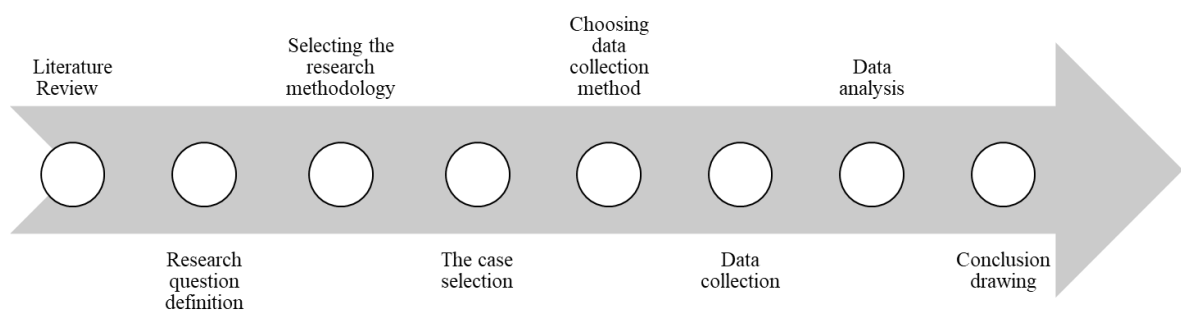


Figure 7. Case study research process phases (adapted from Kähkönen 2011, 32)

A case study approach can be justified in this study as it allows for gathering in-depth interview data that answers "why" and "how" questions. By adopting this approach, the study can delve into the underlying nature of issues and fundamental factors that shape the phenomenon under investigation, thus gaining a more profound understanding of the subject matter. (Kähkönen 2011, 33) The case study approach is appropriate when the research questions aim to clarify contemporary circumstances. It is also a suitable approach when a more in-depth and extensive explanation of the phenomenon is required. (Yin 2018, 33) A case study has also been identified as a suitable research method in supply management, purchasing and supply chain management, as it allows the collection of rich and profound data which can be used to form an in-depth analysis of the phenomenon explored. (Kähkönen 2011, 31)

A case study can be either a single-case or multiple-case study based on how many cases engage in the study. If the researcher has the opportunity to choose and has the appropriate resources, multiple-case designs should be preferred. Two or more cases provide analytical benefits that might be significant. (Yin 2018, 75; 98) Even though a multiple-case study lacks the depth that can be obtained from single-case studies, it provides advantages such as smaller observer bias and the ability to conduct a cross-case comparison (Kähkönen 2011, 35). In this study, data and findings are collected from several companies. Thus, it can be identified that the study follows a multiple-case design. In conclusion, the multiple-case study approach is appropriate for this study.

3.3 Data collection

For this study, the research data was collected through interviews. In case studies interviews have been identified as one of the primary sources to gather data and source material for a study (Yin 2018, 161). Therefore, interviews have been an extensively popular method to gather research data in qualitative research (Eriksson & Kovalainen 2008). Interviews can be structured, semi-structured or unstructured depending on the aim and purpose of the study (Saunders et al. 2016, 390–391). In qualitative research, interview questions are usually but not always unstructured and open-ended as the aim of the research is often to elicit interviewees' views and opinions (Creswell 2014, 294). The interview structure typically in

a case study follows a guided structure instead of relying on structured queries (Yin 2018, 161).

In this study, interviews were conducted as semi-structured interviews. This approach covers themes and key questions related to the study (Saunders et al. 2016, 391). The advantage of this interview approach is that it is not as strict as a structured interview. Semi-structured interview approach can also bring out information that might not have been considered if answers had been provided to interviewees in advance. (Tuomi & Sarajärvi 2018) According to Eriksson and Kovalainen (2008), in a semi-structured interview, the researcher can take a prepared set of topics, issues or themes but still maintain the ability and freedom to change the order of questions and alter the interviewee's wording in each interview. The tone of the interview can remain informal and conversational, but the interview materials remain systematic and comprehensive (Eriksson & Kovalainen 2008).

Especially in business research, qualitative interviews are often done as semi-structured interviews as they allow for a more in-depth exploration of topics and better helps the researcher identify patterns and themes, allowing for a more comprehensive understanding of the phenomenon (Eriksson & Kovalainen 2008). For these reasons, it can be concluded that a semi-structured interview is suitable for this study.

3.4 Case selection

The selection of cases for this study was based on a comprehensive set of criteria. The initial selection of cases was influenced by the fact that the study was commissioned by a Finnish grocery retailer. Therefore, the Retailer was purposefully included in this study and was an active participant in the case company selection process. Five suppliers were chosen to participate in the study in addition to the Retailer. The suppliers were chosen on three main criteria. The first criterion was that the company is Finnish and operates in the food retailing sector. The second criterion was that the company has existing operations with the Retailer. The third criterion was that the company manufactures and supplies fresh perishable products. Essential for this study was to make sure that a comprehensive and versatile set of companies is included in this study to represent the Finnish supplier landscape in fresh food retailing.

After the supplier companies had been selected, additional criteria were established to choose the correct interviewee from each company to participate in the study. Individuals were chosen on the basis that they have extensive understanding and vast knowledge about their company's supply chain operations and supply chain management. Another criterion emphasized in the interviewee selection was that individuals have knowledge about food waste management in food supply chains and know about their own companies' food waste management practices. The interviewee from the Retailer company was chosen on the same criteria as those from the supplier companies. Additional criteria for the interviewee from the Retailer company was that the individual has experience with working the suppliers selected for this study. The interviewees and companies are presented in this thesis in such a way that they cannot be identified. This is done to protect their anonymity, as well as to protect the companies' trade secrets. Interview details are presented in Table 1 below.

Table 1. Interview details

Company	Description of the primary products	Position of the interviewee	Duration of the interview
Supplier A	Fruit- and vegetable products	Transport Manager	73 min
Supplier B	Bread- and bakery products	Supply Chain Director	49 min
Supplier C	Meat, processed meat and convenience food	Planning Manager	59 min
Supplier D	Meat, processed meat and convenience food	Head of Demand Management	64 min
Supplier E	Dairy products	Director	48 min
Retailer	Grocery retailer	Supply Chain Director	65 min

The case companies were initially contacted by a manager from the Retail company. The interview request was sent to five different fresh produce suppliers that supply the Retailer. All five suppliers accepted the request to participate in the study. After this, interview dates were scheduled with each supplier and also with the individual from the retailer company. Interviews were conducted in January and February 2023. All of the interviews were done via Microsoft Teams except for one which was done in-person. The length of the interviews

varied from 47 to 73 minutes. Consent to record the discussion was asked of each interviewee. The ability to record audio data is helpful when later reciting and analysing the data. After the interviewees, each interview was transcribed into written form to aid in the analysis process.

3.5 Data analysis

The objective of data analysis is to extract significant insights and draw conclusions pertaining to a particular subject matter, through the development of a thorough and well-founded depiction of the phenomenon under investigation, based on the available data (Puusa et al. 2020). This study used a content analysis approach to analyse the collected data. Krippendorff (2019) define content analysis as “*a research technique for making replicable and valid inferences from texts (or other meaningful matter) to the contexts of their use.*” Content analysis often involves a systematic process of collecting, analysing, and interpreting data to draw conclusions (Hsieh & Shannon, 2005). The content analysis approach is one of the most commonly applied approaches for examining qualitative data, as it can be used in a wide variety of qualitative research (Puusa et al. 2020).

One of the first steps in the data analysis process is transcribing the interviews into a written form. Transcribing interviews helps the researcher to prepare and organize the data for analysis (Creswell 2014, 247). The data was read over as soon as the interviews had been transcribed. Following the initial readthrough, a more in-depth rereading of each interview was done. The next step of the analysis process was to start coding the data. Eriksson and Kovalainen (2008) define coding as a process where “*the features, instances, issues and themes in empirical data are classified and given a specific label, a code.*” The coding process is usually done to form themes and categories for later analysis. For a qualitative study, the coding process typically reveals five to seven themes or categories that display the study’s major findings. (Creswell 2014, 249) After the research data has been coded, conclusions can be drawn about the phenomenon being studied (Puusa et al. 2020).

3.6 Reliability and validity

The reliability and validity of the study were assessed to ensure the accuracy of the findings. This was done by considering factors such as the sample selection and data collection process. Reliability and validity are the principal factors in business research when evaluating research (Eriksson & Kovalainen 2008; Kähkönen 2011, 38).

Reliability indicates the extent to which the research can be repeated to achieve equivalent results (Eriksson & Kovalainen 2008). It aims to ensure that the study can be repeated in future studies with the same procedures, and similar results and conclusions can be drawn (Eriksson & Kovalainen 2008; Yin 2018, 82). Yin (2018, 82) acknowledges that, in reality, the ability to repeat a case study and achieve comparable results to the original study is rare. Therefore, reliability should be approached so that the study's procedures are as explicit as possible (Yin 2018, 82).

One way to increase a case study's reliability is to form a case study database. This means documenting the study's critical elements, such as the stages of the research process, interview questions, and related procedures (Kähkönen 2011, 39). Puusa et al. (2020) point out that reliability cannot be expressed through objective and quantitative measures in qualitative research. Reliability is always examined in the context of the research and is evaluated depending on the qualitative methods used in the research (Puusa et al. 2020). In this study, efficient reliability is achieved by explaining in detail how the study has been conducted. The interview questions are also displayed in the study's appendices (Appendix 1), meaning that the same questions can be utilized and repeated in future studies. Although, due to the high level of context dependence, it might be challenging to replicate the findings made in this study in future studies.

According to Eriksson and Kovalainen (2008), validity in academic research refers “*to the extent to which conclusions drawn in research give an accurate description or explanation of what happened.*” The correctness of the measures, accuracy, and generalizability of the findings are considered when analysing the study's validity (Saunders et al. 2016, 202). An essential aim of validity is to ensure that the study is accurate and correct (Eriksson & Kovalainen 2008). In a case study, validity can be assessed by evaluating the study's

construct-, external- and internal validity. Internal validity is only evaluated in explanatory or causal studies. (Yin 2018, 78–79) As this study is exploratory internal validity is not taken into consideration.

Construct validity aims to identify that suitable operational measures are applied in the study. For example, construct validity can be increased by gathering data from multiple sources and allowing interviewees to review the case study report. (Yin 2018, 78–79) This study was conducted as a multiple case study to increase the construct validity of the study. Case companies were chosen from different backgrounds to improve reliability. Additionally, different sizes of case companies were chosen to participate in the study, and both large, medium, and small companies were interviewed. Furthermore, the interviewees and study examiners reviewed the study findings before publication to further increase the construct validity of the study.

According to Yin (2018, 351), external validity can be defined as *“the extent to which findings from a case study can be analytically generalized to other situations that were not part of the original study.”* Typically, in qualitative research generalizability is not a significant element. Generalizability in qualitative research is rarely achieved because studies often focus on a unique event or set of people, making it hard to utilize the study's findings broadly. For this reason, in qualitative research, generalizations usually focus only on elements such as having similar kinds of people, times and settings to the initial study. (Johnson 1997 289–290) This study provides information about the individuals and the number of people interviewed to achieve a sufficient level of external validity. However, preserving the anonymity of the individuals and their companies limits the information provided. External validity is also further achieved as it has been described as the criterion that the companies and individuals have been selected for the study. Overall, it can be concluded that several steps were taken to ensure and increase the study's validity.

4 Empirical Findings

This chapter presents the empirical findings of the study. The first subchapter provides an overview of the role of supply chain management in the case companies. The second subchapter introduces how the case companies have implemented collaborative supply chain practices and their views towards them. The third subchapter covers food loss and waste and how the case companies address this issue. Finally, the fourth subchapter combines the topics of supply chain collaboration, supply chain integration, food loss and waste management.

4.1 The role of supply chain management in the case companies

At the beginning of the interviews, the study aimed to understand the role of supply chain management in the case companies and how it is perceived. According to Supplier A, supply chain management has a critical role in their operations as they manufacture and sell "ultra-fresh" perishable products with short best-before dates. The Supplier stressed the importance of efficient supply chain management to ensure that products reach retailers' stores and that retailers receive products with sufficient best-before dates. Likewise, Supplier B highlighted the significant role of supply chain management in their company, particularly in terms of good forecasting to meet the extremely short timeframe from order to delivery that the company operates with. Overall, both interviewees agreed that supply chain management plays a central and crucial role in their companies.

Supplier C pointed out the importance of supply chain management in achieving the excellent delivery reliability that their company is known for. The interviewee explained that the supply chain drives both their production and demand planning. On the other hand, Supplier D recognized that while supply chain management plays a significant role in their company, there is still room for improvement in terms of supply chain practices and management structure. Supplier D manages a complex network of people and practices, which demands a multistep approach to supply chain management. Despite the involvement of numerous parties, the Supplier strives to have complete control over the entire supply chain. Supplier E also highlighted the critical role of supply chain management in their

company, particularly for customer satisfaction and operational efficiency. As they also deal with products with extremely short best-before dates, supply chain management is essential to their everyday operations and has been raised as one of the company's main priorities.

The Retailer stated that the role of supply chain management is ever-increasing in their company. The company has moved towards centralizing control of supply chain operations and management in recent years. The company has aimed to integrate all supply chain stages into one centrally controlled system, including elements such as ordering, delivery, and invoicing. The Retailer recognized that continuous supply chain improvement is necessary for achieving better overall efficiency in their operations, particularly as cost pressures in supply chains have recently increased. Additionally, the Retailer aims to develop supplier collaborations, automatic replenishment, and fulfilment practices to achieve higher-quality operations. The interviewee sees the role of supply chain management growing and becoming more critical in the future.

In summary, the interviewees from the case companies agree that supply chain management plays a crucial role in their operations, particularly when dealing with fresh products and products with short best-before dates. The interviewees highlighted the importance of good forecasting, supply chain practices, and management structure to achieve delivery reliability, customer satisfaction, and operational efficiency.

4.2 Supply chain collaboration and integration

For the context of the study, it was important to understand if companies have supply chain collaboration and integration practices in their supply chains. If such practices are in place, it is valuable to understand what kind of practices they have implemented in real-life situations with other supply chain members as well as to understand how they perceive and understand the concept of supply chain collaboration and supply chain integration as well as if they see value in SCC and SCI.

4.2.1 Supply chain collaboration: procedures and perceptions of the case companies

As the case companies were interviewed in regard to their supply chain practices, it became evident that they all have some form of collaboration in place, although the level and practices used varied (Table 2). Supplier A reported that it does engage in supply chain collaboration to varying degrees, including supply forecasting and demand planning with major retailers. The Supplier emphasized that effective collaboration requires communication with supply chain members and that they maintain a good dialogue with retailers as needed. Collaboration between Supplier A and the Retailer is particularly emphasized during peak seasons and holidays, and open communication is essential for finding mutually beneficial solutions. In some cases, the Retailer and Supplier A may disagree on future demand forecasts, necessitating two-way communication to reach solutions that work for both parties. Supplier A also has supply chain collaboration practices in place with its suppliers, though the level of collaboration varies depending on the supplier's size, partnership type, and product type delivered. Additionally, Supplier A noted that their high weather- and seasonal-dependent raw materials make forecasting and planning challenging for the required quantity and timing. Therefore, the interviewee furthermore emphasized the importance of dialogue and communication to stay informed about the raw material situation for their end products.

Supplier B also stated that they have supply chain collaboration practices in place. Supplier B regularly meets with major retailers to discuss further development projects and their status. Supplier B also receives demand forecasts from retailers regarding expected customer demands for their products in the upcoming weeks and months. Additionally, the company provides retailers with information about any possible challenges and upcoming special situations. Supplier B emphasized the importance of supply chain collaboration, particularly due to recent world events such as the COVID-19 pandemic and the Ukraine War. The company has proactively focused on enhancing supply chain collaboration to respond to upcoming changes and challenges that may impact their supply chains and alter product demand. Supplier B highlighted that the current uncertainty in markets and the world has intensified customer cooperation.

Supplier C has also implemented supply chain collaboration practices. According to the interviewee, regular supply chain collaboration meetings are held with major retailers, and these meetings are identified by Supplier C as one form of SCC. Interviewee C, who is responsible for Supplier C's demand planning, further stated that their collaboration practices include a dialogue regarding demand planning, especially with retailers. The interviewee also cited campaign and seasonal analyses conducted via email communication with retailers as examples of collaboration practices. However, the interviewee acknowledged that it can be challenging to distinguish between supply chain management collaboration practices and the commercial side of their operations and collaboration. Supplier C stated that their supply chain collaboration is mainly operational rather than strategic.

Like Supplier B and C, Supplier D also has regular supply chain meetings, which the interviewee D regards as a form of collaboration. These meetings are held jointly with Supplier D and their retail customers. Supplier D emphasized that they communicate the most frequently with the Retailer's replenishment planner, who is responsible for day-to-day supply chain operations. This communication is considered by Supplier D as an operational supply chain collaboration.

Supplier E has established supply chain collaboration practices similar to other case companies. However, the company intends to enhance its collaboration practices to adopt a more structured approach, primarily with retailers. Supplier E's collaboration practices currently rely on interpersonal communication between its personnel and the retailers' supply chain contacts. The company underscores the importance of communication in facilitating its supply chain collaboration. Supplier E communicates daily, weekly, and monthly with the Retailer, suppliers, and other supply chain partners. The company is also actively researching and developing strategies to improve the depth of its supply chain collaboration with retailer customers.

Table 2. Identified collaboration practices by case companies

Case Companies	Collaboration Practices
Supplier A	• Supply forecasting and demand planning with major retailers • Communication and information sharing with supply chain members
Supplier B	• Regular meetings with major retailers • Providing retailers with information about challenges and special situations • Receiving demand forecasts from retailers
Supplier C	• Regular supply chain collaboration meetings with major retailers • Dialogue regarding demand planning with retailers • Conducting campaign and seasonal analyses with retailers
Supplier D	• Regular supply chain meetings with retail customers • Communication with the retailer's replenishment planner
Supplier E	• Supply chain meetings with the Retailer • Research and development of strategies to improve supply chain collaboration with retailer customers
Retailer	• Regular meetings with key suppliers • Operational collaboration practices with suppliers • Supply chain collaboration with replenishment planner and planning organization of key invoicing suppliers

The Retailer noted that they have established supply chain collaboration practices divided into upper and lower-level practices. The interviewee mentioned that upper-level collaboration practices include regularly scheduled meetings with key suppliers approximately three times a year. During these meetings, current supply chain topics are discussed, key performance indicators (KPIs) are reviewed, and future development areas are identified. The Retailer also engages in supply chain collaboration with other suppliers, although the frequency and personnel involved vary. More general-level operational collaboration practices are also conducted with suppliers, such as sharing private label forecasts through a supplier portal. However, this only applies to suppliers of private-label products. The Retailer engages in regular collaboration with both their replenishment planner and the planning organization of their suppliers to forecast seasonal and campaign demand, ensure adequate product availability, and manage food waste. However, this

collaboration is mostly only done with key invoicing suppliers. For suppliers who deliver their products to the Retailer's warehouse, collaboration relies more on personal communication between the individuals responsible for replenishing the specific product group.

4.2.2 Supply chain integration: procedures and perceptions of the case companies

As part of the interview, an essential question was presented to all case companies concerning supply chain integration practices and procedures, similar to supply chain collaboration. Supplier A responded that they do not have supply chain integration practices in place and classified their practices more towards supply chain collaboration. However, the interviewee mentioned that they have certain suppliers with whom they collaborate more in depth, which could be considered to have some elements of integration. One aspect highlighted by Supplier A was that the Retailer offers an information-sharing service for a fee, providing various sales and reporting data. Supplier A believes that this service can be considered an integration element. However, the information service is only used by the commercial side of Supplier A, and the information is not applied to their supply chain operations and management.

Like Supplier A, Supplier B also stated that they do not have supply chain integration practices. The reason for not having integration practices is partly due to the belief that Supplier B feels that they have a better understanding of demand than their retail customers. As a result, integration is considered to be somewhat unnecessary. However, Supplier B acknowledged the need to shift towards an approach where shared information and data drive the entire supply chain. This method of operating can be regarded as a form of integration according to Supplier B. Despite Supplier B sharing a significant amount of information with the Retailer to promote operational cooperation, the interviewee perceives this as a collaboration rather than integration.

Supplier C was the first case company to be interviewed where supply chain integration practices had been established. Supplier C receives information from retailers that is either manually or automatically entered into their demand planning software, depending on the situation. This data is then compared against forecasts made by retailers to identify any

differences or potential challenges. In addition, Supplier C highlighted the use of automatic exception reports, which are generated using both supplier and retailer data to ensure proactive decision-making. However, the demand forecasting process only covers a small set of retail stores, and Supplier C feels that there is still room for improvement.

Supplier D stated that their company has a more in-depth collaboration with the Retailer that could be identified as a type of integration. However, Supplier D struggled to differentiate between collaboration and integration in their supply chains and management. Supplier D mentioned that they work closely with the Retailer's replenishment planner. Together with the replenishment planner, they plan campaigns and seasonal operations to maximize shelf availability for their products. Although, the interviewee was not available to categorize if this was a type of integration or simply a form of collaboration.

Supplier E also pointed out the challenge of categorizing collaboration and integration. The interviewee noticed they have some integration with individual products sold by the Retailer, but the interviewee would not necessarily categorize it as true integration. Supplier E noted that they have integration practices with their key material suppliers where the traditional supplier-buyer approach has been removed, and operations are merged. Supplier E highlighted an example with their packaging material supplier, with whom they have formed a joint team to handle the subject matter in an integrated manner.

The Retailer sees that they do not have supply chain integration practices. The interviewee sees that they have in-depth collaboration examples with previously mentioned key invoicing suppliers, which considers factors such as campaigns, seasonal activities, as well as new product launches. Although the Retailer regards this as collaboration, as the information still relies heavily on personal communication between the Retailer and supplier personnel. The Retailer pointed out that they share data through the earlier mentioned information-sharing service where suppliers can access information such as sales data, store food waste information, and other relevant sales info. Additional data is shared with many suppliers, but the Retailer does not see this as supply chain integration. The Retailer sees that the data is mainly shared from the Retailer to suppliers, and thus, there is a lack of two-way data sharing between the supply chain members. The Retailer is also developing a new

supplier portal where supply chain information is shared more with the suppliers, but this is also seen as a type of collaboration instead of integration.

4.2.3 The case companies' perspectives on the role of collaboration and integration in supply chain operations and management

After establishing what kind of collaboration and integration practices the case companies have, the study aimed to understand how the case companies intend to use collaboration and integration in their supply chain operations and management. Supplier A pointed out that their company aims to utilize both SCC and SCI in their supply chain management. It was mentioned that their company intend to utilize a mutual exchange of information and historical data about how the products have been sold in stores in order to make sure that their products are in stores at the right time so that stores have products that can be sold, and the end customers have goods that they can buy. According to Supplier A, the use of SCC and SCI practices enables achieving these goals.

Supplier B also acknowledged that SCC and SCI are intended to use in their supply chain operations and management. Supplier B explained that collaboration and integration are aimed to ensure that they have raw materials, packaging materials and the correct amount of capacity in their production at the right time. In a short timeframe, Supplier B pointed out that their company aims to utilize collaboration and integration to enhance their forecast accuracy so that production surplus can be reduced and eventually eliminated. Food waste and surplus production management are emphasized by Supplier B as areas in which they intend to use SCC as well as SCI.

Supplier C mentioned that they intend to use supply chain collaboration and integration to give their production as accurate as possible information regarding the demand from their customers, otherwise known as the retailers. This is further emphasized during campaigns and seasons so that the Supplier can guarantee their customers the required amounts while minimising their internal waste. Supplier C mentioned that a decade ago, their demand planning for campaigns was mainly done on an ad hoc basis making waste management a challenge.

Supplier D, like other suppliers, stated that they intend to use SCC and SCI in their supply chain management operations. Supplier D recognized that utilization of collaborative practices has, in past years, enabled them to achieve better results as Supplier D has understood better how the Retailer's SCM operations work. This has resulted in a better understanding of how the Retailer removes products from their product selection and the timeframes and dates when they change their replenishment planning. Supplier D pointed out that this kind of utilization of collaboration will enable them in future to achieve better results in their operations, mostly related to demand planning.

Supplier E wants to see that in the future, collaboration and integration in retail supply chain management enable them to increase dialogue between supply chain partners, thus forming an opportunity to recognize products that form food waste at the end of the supply chain. Supplier E wants to see that information regarding the Retailer's warehouse and store balances, such as stock levels and inventory data, would be available to them. Based on this information, they could better balance their production and ensure they produce the right amount of goods at the right time. This would enable significant savings to be made in their operations and benefit the entire supply chain. Supplier E mentioned that they would see vendor-managed inventory (VMI) as a good way to improve further the performance of not only their operations but the whole supply chain that considers all of the supply chain members.

The Retailer also sees the potential for further developing its operations through collaboration and integration. The interviewee emphasized that such activities should be done in cooperation with the suppliers to achieve the desired goals. The Retailer underlined that SCC and SCI require considerable resources and work effort. Therefore, they must be built on a solid plan that ensures the investment of resources and time in such activities produces positive results. Additionally, the Retailer pointed out that it is crucial to evaluate which suppliers are potential partners for collaborative and integrative efforts. The main benefits of SCC and SCI for the Retailer include securing supplier delivery reliability and ensuring product availability for stores. At the moment, the Retailer is utilizing collaborative practices to the extent that resources allow. However, in the future, they would like to see more integration in their supply chains on a broader scale to achieve more automatic and transparent supply chains.

4.2.4 Value in supply chain collaboration and integration

All the case companies recognized the substantial value within supply chain collaboration and integration. Supplier A suggested that these practices hold even greater value than presently recognized, with significant potential yet to be harnessed. In Supplier A's view, all supply chain members stand to gain improved business results in the long term, despite potential short-term losses. Similarly, Supplier B noted that SCC and SCI constitute as crucial components in supply chain value creation. The interviewee emphasized that achieving genuine value creation requires the participation of all members in collaborative efforts.

Supplier C is confident in their ability to utilize SCC and SCI to create value owing to their effective resource and demand planning systems. However, the Supplier acknowledges that in order to optimize the benefits of collaborative practices fully, it is necessary to understand the thought processes and operational procedures of the individual replenishment planners at the Retailer. Supplier C believes they can further enhance their supply chain integration practices and improve overall performance by gaining a deeper understanding of how these planners operate.

Supplier D acknowledges the substantial value that collaborative efforts bring to their company's supply chain management and overall business performance. In addition, Supplier D recognizes the importance of internal cooperation in achieving supply chain goals, which generates significant value for the organization. Supplier E likewise considers SCC and SCI as vital elements in terms of value creation. According to the interviewee, collaborative practices can enable them to optimize their production batches, reduce waste, and achieve substantial monetary savings.

The Retailer regards collaboration and integration in the supply chain as crucial factors for achieving future success. The interviewee from the Retailer company views the SCC and the SCI as essential in creating more integrated, transparent, and manageable supply chains. Consequently, the Retailer recognizes the considerable value and potential of these practices for driving positive business outcomes.

4.3 Food loss and waste management in the case companies

The second dimension examined in this study was the management of food loss and waste in the case companies. A set of questions was posed during the interviews to the case companies to examine their approaches and actions concerning FLW. Supplier A expressed their commitment to managing FLW to the best of their abilities. Their strategy involves sending end products to retailers as soon as possible to reduce the time products spend in storage, which helps minimise food waste risk. Due to the nature of their raw materials, Supplier A experiences some inevitable food loss that is unsuitable for human consumption. FLW management is also emphasized in Supplier A's company values and strategy, even if they are not documented in writing.

Supplier B expressed that the significance of food loss and waste management extends to their entire organization, with the company establishing substantial responsibility goals for the entire group. As a result, the role of FLW management has become a heightened priority. Supplier B has set an objective to reduce food loss and waste generated by 50% by 2027. The interviewee recognized the inherent difficulty in meeting this objective, requiring essential practice changes and cooperation with retailer customers and suppliers. Supplier B emphasized that enhancing supply chain operations alone is insufficient, also necessitating a revision of other areas, such as category management and product development processes, to manage FLW effectively. Additionally, Supplier B has comprehended that FLW management is a universal challenge for all business units rather than being limited to supply chain management. FLW has also been set as an indicator impacting their employee bonus program.

Supplier C underscored the significance of food loss and waste management in their company, driven by their commitment to sustainability and responsibility goals. The interviewee highlighted their efforts to reduce waste and develop practices to minimise FLW creation. As stated by the interviewee, the company's objective is to minimise food loss and waste since it leads to financial losses and is considered irresponsible.

Supplier D placed significant importance on food loss and waste management. The company regards it as a crucial matter and takes it seriously. The company consistently strives for

operational excellence, including minimising the formation of FLW. FLW is a key indicator utilized in Supplier D's supply chain management. The interviewee emphasized that different perspectives on FLW management exist depending on where the waste is generated. The interviewee highlighted that all areas, such as raw material waste, packaging material waste, and end product waste, should be monitored and measured regardless of where the waste occurs. Supplier D recognizes the role FLW management has in sustainability and the value it brings to society and customers. Therefore, they want to emphasize waste management and make it a more substantial part of their operations. However, the interviewee also acknowledges that there is room for development in the operational sense.

Supplier E places food loss and waste management within their sustainability agenda, which is integral to the organization's operations. The company aims to minimise FLW to meet the climate objectives established for the organization. Like Supplier B, Supplier E aims to cut their food loss and waste in half. However, Supplier E did not provide a specific timeline for achieving this objective but emphasized the need for rapid action. Due to the perishable nature of their products, the role of FLW management is vital for Supplier E. The interviewee underlined that FLW management is an essential matter in terms of profitability.

The Retailer acknowledged that FLW management plays a significant role in their company and is a key performance indicator that guides their operations. In line with their sustainability strategy, the Retailer has announced their intention to reduce their waste by half, similar to Suppliers B and E. This goal is set to be achieved by 2030, using the food waste level of 2019 as a benchmark. The Retailer emphasized that reducing FLW is a strategic priority, considering both sustainability and business factors as key aspects of the issue. The Retailer takes a pragmatic approach towards the issue, recognizing that zero waste is not a realistic goal for their business. The interviewee stressed the need for waste management to be conducted in a controlled and responsible manner.

4.3.1 The evolvement of food loss and waste management in the case companies

One of the noteworthy parts of the study was to understand how the role of food loss and waste management has developed in the case companies (Table 3). During the interview with Supplier A, it was noted that global trends have had a significant impact on perceptions

of FLW. The interviewee acknowledged that there has been a growing global attention to the issue, which has resulted in more emphasis being placed on reducing FLW across supply chains. However, Supplier A did not specify that the FLW management role would have drastically changed in their company.

Supplier B highlighted that the priority of FLW management has increased over the years. The interviewee noted that their company has learned to manage waste in the entire value chain, not just limited to their operations. Moreover, it was pointed out that the escalating prices of raw materials and energy have further emphasized the importance of FLW management in the supplier's company. In addition, the role of sustainability goals has been better comprehended in the company, which has further amplified the role of FLW management. Supplier B noted that the thinking has shifted from minimising waste in a single supply chain area to minimising waste in the entire chain.

According to Supplier C, the interest in measuring FLW has increased, resulting in its heightened importance. However, the interviewee did not recognize any significant changes in the company's views towards FLW management. Food waste has been an area of interest for Supplier C for the past two decades, but better data utilization has enabled better access to the root causes. Through improved forecasting and a systematic approach to operations, lower levels of waste generation have been achieved.

Supplier D acknowledged that they are able to handle food loss and waste more efficiently than before, although the interviewee did not observe any significant changes in attitudes towards the subject. Waste management has been integrated into their operations already in the past, similar to Supplier B. However, Supplier D recognized that global challenges and trends have likely increased the priority of the subject in the last few years. The interviewee sees a positive development in the subject resulting from enhanced cooperation between internal departments, joint planning, better communication, and an improved ability to anticipate the future.

Table 3. Approach towards FLW management in case companies

Case company	Internal approach to FLW management	Market approach to FLW management
Supplier A	Remained the same	Increased
Supplier B	Increased	Increased
Supplier C	Remained the same	Increased
Supplier D	Remained the same	Increased
Supplier E	Increased	Increased
Retailer	Increased	Increased

Supplier E stated that food loss and waste management has become more prominent nowadays than in the past, even though it has been an integral part of their operations also in the past. The interviewee highlighted that the current situation is a challenge for their company, as rising inflation reduces the amount of money that end customers have, resulting in an increased emphasis on minimising food waste. This situation puts pressure on Supplier E as a supplier, as well as on the retailer stores, with regard to FLW management, as the sales volumes decrease.

The Retailer also observed that the role of FLW management has increased in their company. The interviewee highlights two primary factors that have strengthened the role. Firstly, waste impacts profitability, and as the Retailer has gained awareness of waste management and ways to report waste, the interest in FLW management has increased. Secondly, the role of sustainability has gained importance over the years. This increased emphasis on sustainability has resulted in a greater interest in FLW management, which according to the Retailer, will continue to increase in the future.

4.3.2 Food loss and waste management in supply chains and as a part of supply chain management in the case companies

The case companies were asked about the link between their supply chain management and food loss and waste. Additionally, they were asked to elaborate on how their FLW management practices align with their overall supply chain strategy. Supplier A emphasized the significance of an efficient, streamlined, and swift supply chain that aims to prevent food waste production. The company's objective is to continuously optimize their supply chains and management practices to prevent FLW in their operations. The overall aim of their supply chain management with regard to FLW is to minimise the process time and deliver products to end customers as fresh as possible, thereby reducing the risk of FLW.

The relationship between FLW and Supplier B's supply chain management is evident in the company's key performance indicators. Delivery reliability is emphasized as the most crucial KPI, while food waste is considered the second most significant KPI. Although the interviewee did not specify how FLW management is incorporated into the company's supply chain strategy, it was emphasized that FLW management is included in their long-term planning. In addition to food waste, other sustainability criteria, such as energy efficiency and carbon dioxide emissions, are also considered in this planning. Supplier B stresses that food waste is always considered when making investments in their supply chains, and the impact of these investments on the level of food waste generated in the company's operations is evaluated. The company is firmly committed to sustainability goals and applies pressure on its supply chain management and the entire organization to reduce the amount of food waste. Furthermore, Supplier B recognizes that food waste has monetary value and that better FLW management in their supply chains contributes to better business outcomes.

The interconnection between FLW and Supplier C's supply chain management is reflected in the company's FLW measurement. Supplier C has emphasized the importance of minimising FLW in their supply chain operations, and the company has set objectives to measure and track FLW reduction. The interviewee emphasized that the company receives increasingly strict FLW reduction targets, and their goal is to decrease FLW generation annually. Supplier C recognizes the significant role of production and production planning

managers in reducing FLW in their overall operations. While discussing the strategic role of FLW in their supply chain, Supplier C identified FLW as a more integral part of their sustainability strategy than their supply chain strategy. Although the interviewee acknowledged the critical role of FLW in their supply chain operations, they recognized that the role is continuously increasing, even though it is not explicitly included in their supply chain strategy.

Supplier D considers the relationship between FLW and their supply chain management to be of significant importance. The interviewee revealed that they recently reorganized their order-delivery process organizational structure and updated their goals and performance indicators, with FLW being highlighted as one of the most critical indicators for their performance. While FLW management is not necessarily part of the company's supply chain strategy, it is seen as essential to their company-wide strategic goals. Supplier D strives for excellence, promoting change programs within the company that, if successful, could lead to a reduction in the amount of FLW generated not only in their supply chain operations but also throughout the entire company.

Supplier E emphasized the crucial relationship between FLW and their supply chain management due to the low shelf life of their key raw material. The interviewee noted that supply chain management and production planning are critical in ensuring the raw material is not wasted. The Supplier faces a significant challenge as the key raw material is pushed to them by their suppliers. Supplier E also pointed out that supply chain management is crucial in balancing unexpected situations where demand suddenly increases or decreases. FLW management is essential to the company's supply chain strategy for these reasons. Moreover, Supplier E highlighted their supply chain strategy to ensure that they can produce goods that are less likely to turn into waste at the customer level.

The Retailer has implemented measures to improve total waste management driven by FLW reduction, in line with their supply chain strategy. Waste management initiatives have been developed based on how retail stores operate to address any waste that can be better managed and mitigated in stores. These initiatives are then integrated into the Retailer's supply chain management, and the Retailer aims to collaborate with their suppliers to develop operations to ensure retail stores receive fresh goods. As the Retailer believes that FLW is an integral

part of achieving overall efficiency in their stores, it is an essential part of their supply chain strategy.

4.3.3 Food loss and waste management elements and practices in the case companies

Elements and practices place for food loss and waste management

During the interviews, the case companies were questioned about the specific elements and practices in place for food loss and waste management. Their responses were found to be reasonably similar to the earlier answers regarding the relationship between FLW management and supply chain in the case companies as a whole. However, each company had a unique approach to FLW management. Supplier A noted that its FLW management practices can be categorised into three distinct aspects. The first involves ensuring that raw materials are obtained as fresh as possible, and that the circulation of these materials is kept as short as possible. The second element is maximising the usage time for semi-finished products processed from these raw materials. Finally, the third element focuses on expediting the delivery of final products to retailers.

In contrast, Supplier B's approach to FLW management is more strategic, as discussed in previous sections. The company's overarching goal is to minimise FLW in all its operations proactively. Supplier C's FLW management approach focuses on maximising the reuse of materials to minimise waste. The interviewee highlighted the company's efforts to repurpose some materials by reusing and incorporating them into other products. Moreover, supplying food and raw materials to animal feed is viewed as a viable method to reduce waste that would otherwise end up as FLW by Supplier C.

Supplier D's FLW management practices and elements were found to be closely related to their supply chain management and production management practices, identified in previous sections. According to the interviewee, they closely monitor the expiration of their products, and if they identify a risk that some product batch might end up as FLW, they donate it to charity. Additionally, the interviewee mentioned that they adjust prices to sell products and actively circulate them before they go to waste. However, Supplier D emphasised that proactive measures are taken at earlier stages of the chain to prevent such situations.

Supplier E reported that their FLW management practices and elements are primarily determined by the raw materials used in their production process. The Supplier employs a two-stage planning approach involving rough and fine planning to minimise potential losses in their operations. When the risk of FLW is identified, the Supplier engages in discussions with its retail customers to explore ways to prevent it. Additionally, the supplier can accelerate the movement of products themselves to avoid waste. The Supplier also uses a practice called 'dumping' to sell products with a potential risk of generating FLW while still obtaining compensation for them. If the Supplier is not able to sell the products they usually donate them to various charities. The interviewee emphasised that the worst outcome is when products end up as bio-waste.

The Retailer's FLW management practices and elements are closely related to their supply chain management. The interviewee emphasised the importance of setting clear and measurable goals for FLW reduction and identifying the sources of waste generation. Much of their FLW management efforts are focused on their retail stores, which are provided with guidance and training on minimising waste.

Company initiatives for reducing food loss and waste in the food industry

The case companies were also questioned regarding their interest in actively seeking ways to reduce FLW in the food industry. Supplier A acknowledged their involvement in such initiatives whenever possible, focusing on developing solutions to decrease FLW across the industry. The interviewee cited an example of directing raw materials, that cannot be utilized in their own production, to another company capable of using them. In contrast, Supplier B reported for the past four years they have actively been searching for ways to reduce waste in the entire chain, not just within their operations. Supplier B analyses which of their products generate waste in retailer stores and investigates the reasons behind it.

During the interviews, Supplier C was unable to confirm whether their company actively participates in the development of approaches to reduce FLW in the food industry. However, the interviewee highlighted the company's ownership stake in a firm that utilizes food waste for energy generation, which can view as a potential opportunity for developing new waste management methods. In contrast, Supplier D disclosed that their company has appointed a

director responsible for overseeing large-scale and multi-stage initiatives to identify approaches for improving waste management practices in the industry.

The Retailer stated that they are actively engaged in developing ways to reduce FLW in the food industry. For instance, they are about to implement a new supplier portal that distributes up-to-date information to suppliers, enhancing transparency in the supply chain. The interviewee emphasised that their company is not merely reacting to market trends but is proactively developing new ways to mitigate FLW.

Responsibility and accountability in food loss and waste management and development

In relation to the management of food loss and waste, it was essential to identify the parties responsible for the case companies' FLW management and development (Table 4). For Supplier A, FLW management is considered 'everyone's responsibility', with the quality department playing a critical role in ensuring the shelf lives of products and raw materials. However, Supplier A acknowledges that managers and higher-level roles have the power to influence FLW management and development in practice.

Table 4. Responsibility and accountability by case companies

Case company	Responsibility/Accountability
Supplier A	Department driven / Unspecified
Supplier B	Individual in charge / Department driven
Supplier C	Department driven
Supplier D	Department driven
Supplier E	Department driven
Retailer	Department driven

On the other hand, Supplier B has appointed an individual to coordinate their FLW management, as sustainability is a top priority for the company. While FLW management and development are coordinated centrally, Supplier B recognizes that different business units require separate development processes. In the past, Supplier B had not realized that

all business units played a part in FLW management. Therefore, measures and objectives are now aligned across the company's internal units.

Supplier C handles the subject independently through different business units without an individual solely responsible for their FLW management. The production unit is primarily responsible for FLW management and development, although the company has established processes and guidelines. Similarly, Supplier D has no individual responsible for FLW management and development, with the responsibility divided across various business units. On the other hand, Supplier E places the primary responsibility for FLW management and development on their head of production department, who divides the responsibility between quality control and production planning, each with their own KPIs.

The Retailer has not designated an individual specifically responsible for FLW management and development. Ownership of the matter is currently divided across different units, each with their own objectives. While the Retailer has a responsibility department working towards FLW management from a sustainability perspective, they also have a commercial side interested in FLW management's profitability and a supply chain management department responsible for operational procedures to mitigate and control FLW. The Retailer acknowledges the need for more apparent ownership and centralized direction for these currently decentralized operations.

Data and information utilization in food loss and waste management

All case companies identified that data and information are utilized in their FLW management, although to varying degrees. The case companies noted the utilization of data gathered from different performance indicators measuring FLW as one of the key aspects of data utilization in the case companies' FLW management practices. Supplier A highlighted that in their management of FLW, they make use of both their own data and data provided by the Retailer. At the moment, the utilization of data in FLW management is mainly based on computational forecasting and on salespeople's expertise to understand the data. The interviewee identified that artificial intelligence and machine learning might offer opportunities for the company to better utilize data in FLW management and long-term planning.

Supplier B also recognized that data is used in their FLW management, although the interviewee pointed out that data about product availability and waste are obtained with a delay. Information and data are utilized by Supplier B to identify products that pose challenges to FLW management. The interviewee noted that data and information should be used to balance the amounts delivered by the Supplier and the amounts sold out by retail stores, as this could result in overall lower amounts of FLW in the entire supply chain. Supplier B has identified how existing data and information can be used to balance supply and demand but noted that the current challenge is to expand this operation to affect all retail stores that operate differently.

Supplier C pointed out that data allows them to understand how much FLW has been generated. Together with their production planning, they utilize this information to find out further the causes that have resulted in this. The interviewee noted that they would benefit from more detailed information and data from retailers about FLW generated in retail stores, as the Supplier could use this information to understand what has been done right and what has been done wrong in their own operations.

Supplier D struggled to grasp how data and information are used in their FLW management. The interviewee reflected that they receive data and information about if some products have resulted in FLW from the Retailer. With this information, Supplier D aims to discover and research why FLW has been generated and to understand how these kinds of situations could be avoided in the future.

Like Suppliers A, B and C, Supplier E also mentioned that data and information are used in their operations concerning FLW management. They compare their current forecasts to last year's ones to identify possible challenge points. Supplier E is implementing artificial intelligence practices to help further produce better demand forecasts, which could result in better FLW management. However, Supplier E noted that they still are missing large data sets from retailers that could be utilized in FLW management and their operations more broadly. The interviewee noted that the more they have information and data about how the retailer has sold products, for example, during past seasons and campaigns, data about retailers' warehouse balances and forecasts about possible future demands, the better they can plan their production resulting in less FLW.

The Retailer pointed out that they aim to utilize data and information in their FLW management as broadly and comprehensively as possible. The interviewee reflected on earlier indicators about FLW that form their goals and guide them. Numerical goals are placed not only on product group levels but also at store chain levels. Monthly data and information are utilized to analyse how the FLW levels have developed, and this information is also sent to suppliers, most accurately on individual product levels. The Retailer highlighted that data about FLW can, for example, give them information if product lot sizes that should be dropped. FLW data also dictates the Retailers' decisions about removing products that repeatedly result in FLW. Another perspective where the Retailer in FLW management uses data and information is the evaluation of store inventory values and assortment sizes. The Retailer noted that even though data and information are available, the challenge lies in having tools that could be used to make the necessary changes.

4.4 Supply chain collaboration and integration in food loss and waste management

After focusing on supply chain management, collaboration, and integration, as well as food loss and waste management as separate topics, it was essential to combine these subjects into one. The case companies were asked a series of questions that considered all of these subjects to understand how supply chain collaboration and integration are utilized in the management of food loss and waste.

Implementation of supply chain collaboration and integration in regard to food loss and waste management

The interviews with case companies revealed varying degrees of supply chain collaboration and integration in managing food loss and waste. Supplier A, for instance, noted that collaboration practices have been implemented with the Retailer, but these have mainly centred on managing in-store shelf availability and supplier delivery reliability rather than FLW. The interviewee acknowledged that there could be a conflict between maintaining high in-store availability and minimising waste generation.

On the other hand, Supplier B views their FLW and the Retailer's as a single entity, forming the basis for their collaborative practices. They exchange demand forecasts to identify sources and root causes of FLW in the supply chain, considering SCC and SCI for FLW

management to be one and the same. Meanwhile, Supplier C reported not implementing supply chain collaboration for FLW management with the Retailer. The Supplier did not consider their own FLW management practices as a product of SCC. However, they recognized that SCC and SCI could help identify sources and root causes of FLW and enable more effective management of FLW throughout the supply chain.

Supplier E provided an example of how they have proactively contacted retailers' personnel to address any related FLW issues and made necessary adjustments and developments. The Supplier have achieved a high level of integration with their key suppliers, optimizing their own production for stock order cost-effectiveness. According to The Retailer they and their suppliers have jointly agreed on goals for reducing FLW and implemented concrete practices such as supply chain meetings and data sharing, with some pilot-like practices with selected suppliers to analyse which products generate waste in which retail stores. These collaborative practices have integration elements, although they cannot necessarily be identified as an implementation of supply chain integration for managing FLW.

Use of supply chain collaboration and integration to manage and mitigate food loss and waste with retailers/suppliers

The case companies were also inquired if their companies have used supply chain collaboration and integration with retailers or suppliers to manage and mitigate FLW (Figure 8). Responses could be divided into two groups. Suppliers A, B, E, and the Retailer reported that they have indeed implemented collaborative methods to address FLW. The Retailer, in particular, highlighted that collaboration efforts with other supply chain members have been increasing steadily. Supplier B provided an example of a recent product range review where the entire chain's FLW was examined and evaluated. This analysis allowed them to identify unprofitable products that generate losses at multiple stages of the supply chain. Supplier B further explained that FLW analysis was carried out to understand the root causes of FLW and to facilitate collaborative efforts with other supply chain members.

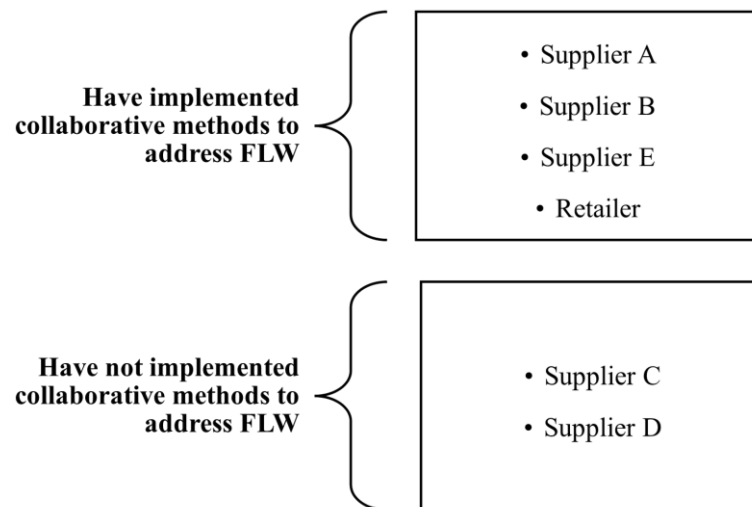


Figure 8. Use of supply chain collaboration and integration to manage and mitigate food loss and waste with retailers/suppliers

Conversely, Suppliers C and D acknowledged that while collaborative efforts were made with the Retailer and other members, these did not specifically aim to manage and mitigate FLW. Supplier C expressed a desire for more information about retailer demand to provide their production with more precise information. They recognized that such information exchange could indirectly reduce FLW by enabling more informed decision-making. Supplier D confirmed that waste management had been discussed in supply chain management meetings, but there has been little collaboration with the Retailer in actively organizing and mitigating FLW.

Perspectives on food loss and waste management in supply chain operations across different stakeholders

During the interviews, the case companies were questioned regarding any possible differences between supply chain members in terms of their perception of FLW management in supply chain operations. Supplier A acknowledged the possibility of variations among members but had difficulty pinpointing any specific differences. However, the interviewee recognized that the current emphasis on sustainability and responsibility in today's world should be reflected in companies' values. Therefore, FLW reduction has gained increased prominence, even though there are differences in how it is managed and mitigated across companies. Supplier A noted that as raw materials progress through the value chain, their

value increases with each value chain part. As a result, the monetary value of FLW rises as products move towards the end of the supply chain, influencing how different supply chain members perceive and approach FLW management.

Supplier B also identified differences in how supply chain members perceive FLW management in supply chain operations. The interviewee emphasized that not all chain members see FLW management as a whole, and some members may only focus on sub-optimizing their operations when it comes to FLW reduction. However, Supplier B sees that supply chain members who consider FLW management as a whole will succeed and achieve beneficial results in the long run.

On the other hand, Suppliers C and D see minimal differences between supply chain members regarding their perception of FLW management in supply chain operations. The interviewees believe that every member aims to minimise FLW and achieve better results in FLW management. Supplier C pointed out that some members may have a better understanding of how to accomplish this. Supplier D highlighted that FLW management plays a crucial role regardless of where in the supply chain the FLW occurs or in what form. The interviewees did not identify any differences in how suppliers and retailers view the subject or any conflicting interests.

Supplier E identified that there are differences in how different supply chain members view FLW management in supply chain operations. The interviewee noted that each company has a different view on the importance of the subject and how it is prioritized in their strategies and operations. For Supplier E, FLW reduction and management is one of the main priorities. However, the interviewee was not certain whether the Retailer shares this same priority as strongly, which may result in differences in how the subject is valued between the two parties.

The Retailer acknowledged that there are significant differences between suppliers in terms of how they perceive FLW management in supply chain operations, leading to differences in how they prioritize and guide their operations. It was noted that some suppliers are more deeply involved in the subject than others, and that the level of interest towards FLW reduction and management varies from one supplier to another. The Retailer emphasized

that supplier maturity on the subject has a significant impact on how the subject is perceived and managed.

The role of information exchange in food loss and waste management.

Based on the interview responses of the case companies, it is evident that information exchange plays a crucial role in food loss and waste management. Supplier A emphasized that without information exchange, all members would work independently, which could lead to inefficiencies in managing FLW. The interviewee further stated that information exchange is a prerequisite for even considering FLW management. Similarly, Supplier B highlighted the importance of information exchange, as it provides facts about FLW and helps in forming a comprehensive picture of FLW in the entire supply chain.

For Supplier C, information exchange plays a role in identifying the root causes of FLW. However, the interviewee highlighted that information exchange is more critical between sales, logistics, and production teams than in demand planning. Supplier C does not consider that there is a systematic information exchange between them and the Retailer to manage FLW at the moment.

Supplier D agreed with Suppliers A and B regarding the critical role of information exchange in FLW management. However, the interviewee from Supplier D emphasized that for information exchange to be beneficial, members must understand each other's business operations and perceptions. Merely pushing information from one member to another is insufficient, as it can result in misunderstandings.

Supplier E pointed out that decisions made without proper information are mere speculations, making information exchange crucial for FLW management and the entire business. Additionally, Supplier E highlighted that more information leads to informed decision-making, resulting in better results for the entire supply chain. The interviewee further emphasized that if FLW reduction is to be achieved in supply chains, information exchange must occur more frequently than it currently does.

The Retailer also considers information exchange to be essential for FLW management. The Retailer stated that they should offer better transparency to suppliers to promote better

cooperation, resulting in a lower total loss in the entire supply chain. The interviewee emphasized the importance of two-way information exchange between supply chain members and automatic information exchange between partners. According to the interviewee, integration is necessary to achieve efficiency and scalability in collaborative information exchange initiatives in the future.

4.4.1 Motivators and benefits of supply chain collaboration and integration in food loss and waste management

From the responses gathered during the interviews, two key factors emerged as the primary motivators for utilizing SCC and SCI in FLW management: business goals and performance. Supplier A emphasized that while their objective is to increase sales and achieve financial growth, they do not aim to achieve this by excessively pushing their products onto retailers, as such actions would result in FLW. The interviewee noted that their company is driven to use SCC and SCI in their supply chains to at the same time offer end customers fresh products with good best-before dates, minimise FLW, and enhance overall business outcomes.

Supplier B also shares a similar motivation for using SCC and SCI in FLW management, as they aim to offer end-customers the freshest possible products, much like Supplier A. Supplier B recognizes that when SCC and SCI are utilized to achieve this objective, the supply of products can be managed effectively, resulting in a successful commercial outcome. On the other hand, Supplier C stated that their motivation for using SCC and SCI in FLW management is primarily driven by growth in turnover and profit. SCC and SCI helps them to increase the accuracy of forecasting that enables the Supplier to make fresher goods and thus lower the risk of wastage.

Supplier D pointed out that their use of SCC and SCI in FLW management is motivated by the ability to generate better forecasts, leading to superior production planning and resource management and, ultimately, reduced levels of FLW. The interviewee noted that one of the motivators of enabling SCC and SCI in FLW management is that by utilizing these practices they are able to boost their key indicator performance. In contrast, Supplier E's motivators differ somewhat from those of Supplier D. Supplier E emphasized that their key motivator

for enabling SCC and SCI in FLW management is to achieve the status of being the most sought-after partner. They are motivated by ensuring that their products do not cause any problems as a waste for retailers, retail stores, or end customers. Moreover, enabling SCC and SCI in supply chain operations allows the Supplier to enhance production efficiency and accuracy, resulting in lower FLW levels.

For the Retailer, the primary motivator for enabling SCC and SCI in FLW management is financial gain. The interviewee acknowledged that all FLW that can be saved translates into monetary savings, which is seen as a means of improving efficiency. The secondary motivators for retailers are sustainability and responsibility goals, but business objectives and monetary savings are the primary drivers for utilizing SCC and SCI in FLW management.

All of the case companies surveyed indicated that the adoption of supply chain collaboration and integration could yield beneficial outcomes in managing food loss and waste. According to Supplier A, greater cooperation in supply chains can lead to better FLW management results. Supplier B suggested that the utilization of collaboration and integration in FLW management helps to reduce suboptimization and fosters greater synergy among members to achieve improved outcomes. The interviewee from Supplier B also remarked that SCC and SCI could help supply chain members comprehend the factors influencing FLW and make necessary adjustments to improve FLW management performance throughout the entire supply chain.

4.4.2 Limitations and challenges of supply chain collaboration and integration in food loss and waste management

The case companies were asked to provide insights on the limitations and challenges associated with using supply chain collaboration and integration in food loss and waste management (Table 5). Supplier A cited trust and the competitive positions of supply chain members as key limitations to SCC and SCI in FLW management. The interviewee emphasized that the competitive positions of retailers can pose a barrier to collaboration as retailers may be hesitant to share information with suppliers who have partnerships with their competitors. Despite the importance of trust and transparency in partnerships, the

Supplier noted that supply chain members often provide only limited information to the other party. Therefore, Supplier A stressed the need for more proactive efforts to foster genuine trust and transparency in partnerships, which are crucial for effective FLW management.

Supplier B highlighted that limitations related to information sharing are a significant challenge to utilizing SCC and SCI in FLW management. According to the interviewee, obtaining information is often difficult and restricted, leading to inefficiencies in managing FLW. In addition, competition laws create another layer of limitation as information needs to be shared equally with all supply chain members. One of the most significant challenges Supplier B faces is the absence of a unified database where relevant information, such as product outselling from stores, FLW, and shelf availability, would be accessible to the Supplier in real time.

Supplier C noted system incompatibility as a limitation and highlighted that although there is interest in collaborative efforts among supply chain members, there may be a reluctance to allocate the necessary resources and time for forecasting cooperation. Meanwhile, Supplier D identified the limited number of products in active supply planning, where collaborative efforts are being utilized, as a key limitation for them, as the Retailer only has a limited number of their products included in the planning.

Table 5. Identified limitations and challenges of SCC and SCI in FLW management

Identified limitations and challenges
• Trust and competitive positions between supply chain members
• Limitations related to information sharing
• Resource limitations
• System incompatibilities
• Over-reliance on SCC and SCI systems
• Cultural change in operations
• Complexity and length of the supply chain

Supplier E identified resources as the main limitation for utilizing SCC and SCI in FLW management, including the number of resources and time, members are willing to allocate towards managing FLW collaboratively. Another limitation identified by the interviewee is the varying levels of motivation or interest towards the subject among the members of the collaboration. Supplier E also highlighted that production efficiency can limit the use of SCC and SCI in FLW management, as suppliers need to maintain operational efficiency to ensure the profitability of their business.

The Retailer has identified several limitations to utilizing SCC and SCI in managing FLW, including resources, information and data systems, and data valuation. In particular, resource limitations constrain the amount of effort that can be devoted to collaborative efforts. Currently, limited working hour resources limit the amount of effort that can be put into cooperation with other supply chain members to manage and mitigate FLW. The Retailer identified automation of data sharing as a potential solution to this limitation, but this is hindered by variations in existing information and data systems, which also require financial contributions for proper development. Another limitation identified by the Retailer is how data is valued. The interviewee noted that as data itself has some value, it can slow down the utilization of collaborative practices in FLW management if the company obtaining the relevant data sees business value in the data and is thus not willing to share it or requires monetary compensation for access to it.

In addition to the above-mentioned limitations, the case companies were also asked about the potential challenges that could arise from utilizing SCC and SCI in cooperation with other supply chain members for managing and mitigating FLW. One noteworthy point raised by Supplier A was that with increasing use of SCC and SCI systems in supply chains, there is a risk of over-reliance on these systems. According to the interviewee, this could lead to a weakened reliance on human competencies and skills. Instead, the interviewee suggested that human competencies and skills should be utilized in conjunction with various SCC and SCI systems to achieve optimal outcomes for managing FLW and the entire supply chain.

Supplier B raised concerns about challenges that may arise when supply chain members focus solely on their own numbers and forecasts, instead of considering KPIs in cooperation with other members. This narrow focus on individual performance may hinder the

development of a shared vision among members and hinder decision-making across the entire supply chain. Moreover, if forecasts and estimates about demand are delayed or inaccurate, this could present a significant challenge to effective decision-making throughout the supply chain. In addition, Supplier B also expressed concern about the potential for misinformation and misrepresentation to occur in a supply chain with many members from different organizations. This could undermine collaborative efforts to manage and mitigate FLW and create further challenges for supply chain management.

Supplier C identified a potential challenge that may arise from utilizing SCC and SCI in collaboration with other supply chain members in the management and mitigation of FLW - cultural change in operations. According to the interviewee, companies may have grown accustomed to their current operations, the KPIs they focus on and evaluate, and how forecasts are formed. Therefore, transitioning to a collaborative approach may introduce doubts and take time, as learning new operation models and creating new processes takes time. The interviewee emphasized that cultural change is a challenge that must be managed effectively to ensure the success of collaborative efforts in FLW management.

Supplier D also recognized challenges that could arise from utilizing SCC and SCI in cooperation with other supply chain members when managing and mitigating FLW. One of these challenges is the sheer number of members in the supply chain, which can make the supply chain more complex and lengthier when collaborative elements are utilized. As a result, the interviewee noted that supply chain control may be lost or become inadequate. This can lead to lost information or information not being directed to the right people at the right time, which can ultimately hinder the effectiveness of FLW management efforts.

Supplier E identified two challenges that may arise from utilizing SCC and SCI in cooperation with other supply chain members for FLW management. Firstly, the implementation of new models, policies, and approaches may lead to short-term or long-term challenges that could impact product availability. Secondly, an unclear division of responsibilities and liabilities among the members may create difficulties when financial losses occur. The interviewee emphasized that these challenges need to be carefully considered and addressed to ensure effective and efficient collaborative FLW management in the supply chain.

The Retailer did not identify any significant risks or challenges related to the subject matter. However, the interviewee agrees with Supplier E's findings regarding balancing FLW management and product availability. The Retailer noted that if FLW management becomes a priority, it may lead to lower product availability levels. Another challenge highlighted by the retailer was the allocation of responsibilities and liabilities in case of failures, i.e., determining who is responsible when failures occur.

4.4.3 Supply chain collaboration and integration in food loss and waste management: achieving best-practice solutions

One of the core elements of the case study was to gather insights and ideas about how best-practice solutions with other supply chain members could be achieved in regard to FLW management. The case companies were questioned about the elements and requirements for achieving these best-practice solutions.

According to Supplier A, open communication, information exchange, and a shared vision among members are crucial elements for establishing best-practice situations. The interviewee emphasized the importance of allowing retailers and suppliers to discuss and compare their decision-making data freely. The objective is to merge information and data from all members to develop a unified view of managing supply chains in regard to FLW management. Supplier A stressed the need to move away from the traditional supplier-customer mindset and adopt a collaborative approach for FLW management and entire supply chain management. The interviewee highlighted that underlying assumptions about hierarchy within the supplier-customer mindset hinder cooperation and act as barriers to achieving true best-practice solutions. Supplier A suggests that having the courage to shift away from these assumptions is crucial for achieving genuine cooperation.

Supplier B noted shared indicators and targets as one of the elements for achieving best-practice solutions with other supply chain members in FLW management. The interviewee emphasized that cooperation and collaborative practices with other chain members should have a clear systematic structure where aspects such as the timeframes and content of meetings would be planned. Another element for achieving best-practice situations according to Supplier B is that operations and projects need to be constantly monitored and

taken forward to have an impact. The interviewee also emphasized the importance of being willing to change the ways of working if no progress is being made.

Supplier C highlighted the significance of improving estimate and forecast accuracy in managing FLW. The interviewee emphasized that achieving 100% accuracy in estimates and forecasts would significantly minimise the amount of FLW generated. In 2018, Supplier C introduced an internal vision that prioritized their demand planning over the retailer's estimations, which would drive the retail store replenishment. Although viewed as a theoretical concept at the time, the interviewee acknowledged that their estimates were more precise at the time than retailers. Thus, this vision was thought of internally as a potential best-practice method to manage demand and reduce the quantity of FLW generated.

To attain best-practice solutions, Supplier D has emphasized the crucial role of trust, respect, and mutual understanding among supply chain members. The interviewee stressed that mutual trust among all supply chain members is a prerequisite for achieving the best-practice solutions that can be developed collaboratively. Furthermore, the interviewee emphasized the importance of establishing strong interpersonal relationships between supply chain members as a means of achieving best-practice solutions.

Supplier E highlighted the importance of mutual interest among supply chain members in achieving best-practice solutions for managing FLW. The interviewee suggested that once mutual interest has been established, members must work towards establishing functional two-way cooperation and exchanging information as much as possible. Sharing new ways of working was identified as another crucial factor in achieving best-practice solutions. The interviewee emphasized the need to incorporate these new ideas into the cooperation. Continuously identifying new elements and ideas through a process of continuous improvement is essential to achieve the best-practice situation in FLW management. Moreover, the interviewee stressed the importance of actively monitoring the level of FLW and delivery reliability in cooperation and making proactive adjustments as needed.

To achieve best-practice solutions in FLW management, the Retailer identified several key elements that must be in place among supply chain members. Firstly, supply chain operations and management must be guided by clear and shared objectives, similar to the perspective

of Supplier B. This ensures that all members are working towards the same goals, which is essential for effective collaboration. Secondly, cooperation between supply chain members must be based on established working collaboration practices before any integration practices can be implemented. This means that members need to have a foundation of trust and effective communication to build on for successful collaboration. This ensures that all members are aligned and committed to the same goals and objectives, which is essential for achieving best-practice solutions.

4.4.4 Future development ideas to supply chain collaboration and integration in food loss and waste management

As a final interview question, the case companies were asked how supply chain collaboration and integration in terms of FLW management could be developed in the future. Similar aspects and elements that were already highlighted as motivators and elements of best-practice solutions were further emphasized as development ideas for the future. Supplier A emphasized that developing SCC and SCI in terms of FLW management requires increased data sharing and transparency among members. The interviewee noted that two different dimensions can be seen related to the subject: the human dimension and the dimension of information technology. Supplier A sees that the more closely these two dimensions cooperate, the more opportunities there are.

Supplier B summarized the areas for future development into three points. The interviewee noted that measurement throughout the supply chain, where both availability and wastage are measured, is needed. Additionally, as noted earlier by Supplier B, supply chain cooperation meetings are needed to achieve systematic control of objectives and operations. As a third point, the supplier noted that members need to better understand different operating practices in the future to understand which goals and objectives require more resources and pose greater challenges than others.

Supplier C noted that greater forecast cooperation and information from customers about what is needed, how much, and when are elements needed to develop SCC and SCI in terms of FLW management in the future. Supplier D could not highlight any specific improvement

ideas. However, it was noted that collaborative practices in terms of FLW management should be developed in the future in cooperation with chain members.

Supplier E identified as a development idea that data and information should be automatically moved across the entire supply chain, giving them the necessary information to guide better how much their retailer customers order and on which days. The interviewee noted that, at the moment, there can be seen two different processes for suppliers and retailers. In the future, SCC and SCI should be utilized to develop one unified chain that reaches the end customer.

The Retailer noted that in order to develop SCC and SCI in terms of FLW management in the future, supply chains needs to become more integrated and coherent. There should not be any separate sections where only a small subsection is optimized. According to the Retailer, good end results are achieved when the supply chain is managed collaboratively as a whole.

5 Discussion

The objective of this study was to investigate the use of supply chain collaboration and integration for managing food waste in food supply chains. In addition, the study aimed to understand how SCC and SCI are perceived and applied by supply chain members as part of their management and operational practices. Moreover, the case study aimed to gain insights into how supply chain members address FLW management. This chapter presents a discussion of the empirical findings obtained from the case study in conjunction with the theoretical framework presented in this study. The purpose of this discussion section is to integrate the empirical findings with the theoretical background in relation to the research objective and questions.

Through empirical investigation, the case companies identified several ways in which SCC and SCI could be leveraged to address FLW management in their supply chains. All case companies aimed to adopt these practices to reduce food waste in their supply chains. The majority had already implemented some collaborative elements in their supply chain management, although not necessarily with the aim to reduce FLW merely.

According to Cao et al. (2010, 6617) and Cao and Zhang (2011, 166), seven key interrelated elements form the basis for achieving supply chain collaboration: information sharing, goal congruence, decision synchronization, incentive alignment, resource sharing, collaborative communication, and joint knowledge creation. The case companies also recognized the importance of some of these elements in their FLW management efforts. In particular, effective collaborative communication and information sharing regarding forecasting, as well as the alignment of incentives through performance indicators, were followed by all case companies. However, while some of the case companies identified goal congruence, decision synchronization, and incentive alignment as critical SCC elements, there was no consensus among them about the importance of these elements.

However, the case companies noted that there is still significant room for development, despite some of these elements already being in place, particularly in retail-supplier partnerships. One of the most significant findings was that supplier companies, in particular,

feel they do not receive sufficient supply chain information and data from retailers. Although some information is shared through established supply chain meetings between suppliers and retailers, it is not considered sufficient. This is a major issue because, as noted in theory, sharing information among supply chain functions and members is one of the main obligations for collaborative inter-organizational relations (Sheu et al. 2006, 27). This could also explain why some suppliers of the case study perceive a clear hierarchical structure between themselves and the Retailer. As Simatupang and Sridharan (2002, 24) have pointed out, information sharing plays a critical role in promoting visibility among members. Additionally, the case companies identified that resource sharing, which covers both tangible and intangible elements (Bergman & Holmqvist 2022, 25), was lacking in the relationship between suppliers and the Retailer.

The case companies emphasized the importance of implementing SCC and SCI in managing food waste in the supply chain. They mentioned various reasons for doing so, such as the exchange of information among supply chain members, enhanced forecast accuracy, better understanding of other members' actions, and improved delivery reliability and product availability. These aspects are critical in reducing food waste and ensuring that fresh products reach consumers in a timely and efficient manner. The empirical findings of the case studies support the theoretical studies by Surucu-Balci and Tuna (2019, 283), which highlight the significance of streamlined, efficient, and fast supply chains in managing perishable fresh products. Thus, implementing SCC and SCI can play a crucial role in reducing food waste in the supply chain, while also improving supply chain performance and enhancing collaboration among supply chain members.

Theoretical studies by Chauhan et al. (2021, 2; 5–6) categorized the reasons leading to FLW into micro-, meso-, and macro-level causes. The empirical findings suggest that most of the reasons leading to FLW in the case companies are related to meso-level causes, which take into consideration issues related to infrastructure and FSC practices. As the study focused solely on Finnish companies, any hard infrastructure-related causes can be eliminated from consideration. Thus, it is assumed that issues in FLW management are mostly related to the case companies' FSC practices. However, some case companies identified infrastructure related to data and information systems as hindering elements, especially for data and

information sharing, forming a barrier to achieving true supply chain collaboration and integration.

Flynn et al. (2010, 59) emphasized that SCI relies on building interactive and close relationships with supply chain members, requiring in-depth resource and information sharing, commonly agreed goals, and synchronized decision making. The empirical results reveal that a unified approach to SCC and SCI is essential for effective waste management, particularly between suppliers and retailers. This finding is supported by Strotmann et al. (2017, 2), who stressed the need for coordinated efforts among supply chain members to achieve a reduction in food loss and waste. Especially since all case companies manage highly perishable products, as noted by Thakkar and Ravalji (2019, 2), the capability to forecast demand is critical. By enabling supply chain collaboration and integration, the case companies noted that they are able achieve better results in their forecasting and demand planning. This is supported by findings from Småros (2021, 5–6), who noted that the ability to notice changes in demand and make necessary adaptations to supply chain actions is more critical when operating with perishable products than with products with long shelf lives that require operational efficiency instead of operational agility and adaptation.

However, it should be pointed out that while SCC and SCI can mitigate and better manage the supply chain in the case companies, there is always likely to be some inevitable waste due to the nature of the products. This finding was also highlighted in the empirical study where case companies identified the nature of fresh products as a significant challenge. This finding is further supported by theoretical findings made by Mena et al. (2014, 151).

For suppliers and retailers, one aspect when establishing collaborative practices as part of their supply chains and waste management is to form a clear understanding of the kind of on-shelf availability percent in retailer stores that is best for reducing the waste generated. The need to calculate and decide the sufficient on-shelf availability is supported by findings by Småros (2021, 27–28), who pointed out that due to the short shelf life of fresh perishable products, 100% on-shelf availability is not feasible.

Bergman and Holmqvist (2022, 29) noted that motivators for supply chain collaboration generally include enhanced knowledge of the supply chain, improved business performance,

and better ability to solve supply chain issues. The empirical results of this study revealed that the case companies identified financial growth, successful commercial outcomes and performance as the primary motivators for utilizing SCC and SCI to manage waste in their supply chains. Furthermore, the companies believed that SCC and SCI could provide an opportunity to reduce waste, ultimately leading to improved business performance, such as increased turnover and profits.

Furthermore, according to the case companies, implementing SCC and SCI could enhance supply chain knowledge and the ability to solve supply chain problems, such as issues related to waste management, thus achieving performance improvements. This aligns with the theoretical findings, as Min et al. (2005, 249) noted that collaboration could lead to improvements in supply chain efficiency, effectiveness, and profitability. Similarly, Ramaunaskas et al. (2021, 711) found that collaboration is associated with better production and capacity utilization, which are benefits that were also identified by the case companies. Additionally, the case companies pointed out that operating with highly perishable products, SCC and SCI practices can help achieve better outcomes when working with these types of products, thus lowering waste levels. This finding is supported by theory, as Shou et al. (2022, 23) identified in their research that integration in supply chains can help achieve more reliable and smoother material flows, which, based on the empirical findings, is critical even when working with highly perishable products.

However, one of the primary challenges encountered during the empirical part of this study was that the case companies often used the terms "supply chain collaboration" and "supply chain integration" interchangeably. While in theory a clear distinction between supply chain collaboration and integration could be made it proved to be more challenging in empirical part of the study. The Retailer emphasized that collaborative practices that rely on personal communication between supply chain members, without the presence of two-way communication, cannot be classified as integration. According to the Retailer, two-way data sharing constitutes one of the crucial elements that define supply chain integration. While a small portion of the case companies understood the differences between the terms the consensus was that the either collaboration and integration were thought as synonyms for each other or especially the concept of integration was not understood in-detail. This finding is to some extent supported by the theory where it was identified that in scientific literature

especially in regard to supply chain collaboration there has been various established definitions (Bergman & Holmqvist 2022, 24). As a result, while we could identify similarities between the theoretical benefits of supply chain collaboration and the empirical findings, it was more challenging to establish similar levels of connection between the empirical findings and supply chain integration theory. Overall, the study found more similarities between the theoretical framework of supply chain collaboration and the empirical findings than with supply chain integration. Despite this, some connections between the benefits identified in supply chain integration theory could still be observed in the empirical findings. For instance, the case companies recognized the reduction of resource waste and better synergy between members of the supply chain as benefits for supply chain integration, which align with the theoretical benefits highlighted in the literature by Alfalla-Luque et al. (2013).

The study's theoretical framework indicates that the benefits of supply chain integration were comparatively fewer than those associated with supply chain collaboration, despite a comprehensive review of the relevant literature. This may explain why companies tend to promote benefits typically associated with collaboration rather than integration. Nonetheless, all the benefits of supply chain collaboration and integration are highly useful in improving food waste management. For example, reduced lead time and increased flexibility, which are benefits of supply chain collaboration (Simatupang & Sridharan, 2002, 23), can promote better waste management in food supply chains. Similarly, more reliable material and information flows and the ability to react to changing market demands, which are benefits of supply chain integration (Småros 2021, 44), can also help improve waste management in food supply chains.

The responses from the case companies revealed that they perceive significant value in SCC and SCI practices, both in their own implementation and as part of food waste management. Although the value discussion did not focus directly on food waste management, it could be inferred that the value recognized in SCC and SCI promotes the companies' interest in implementing these practices throughout their supply chain operations and management while also considering food waste management. For example, the case companies pointed out that SCC and SCI help them gain a better understanding of how other supply chain members operate, which can be considered a beneficial element in managing food waste

generated at the entire food supply chain. The case companies' recognition of the value of SCC and SCI practices aligns with the findings of Chen et al. (2009, 75), who observed that integration helps supply chain members enhance interfirm relationship efficiency and better manage complexity.

When considering the challenges and limitations of using SCC and SCI in food waste management within food supply chains, similarities between theoretical and empirical findings can be observed. In the empirical findings, challenges and limitations were mainly related to resources, individualistic approaches, and information and data-sharing issues. Interestingly, the challenges related to trust were only seen as a potential challenge and limitation to a small extent, despite having been pointed out in theory by several authors in both supply chain collaboration (Bergman & Holmqvist 2022, 30; Benavides et al. 2012; Ramanauskas et al. 2020, 711) and supply chain integration (Katunzi 2011, 108).

In SCC theory, challenges such as a lack of information (Ramanauskas et al. 2020, 711), not sharing real-time data, and focusing on forecasting demand individually (Pierre 2020) were identified as similar to those found in the empirical part of the study. Resource-related challenges were also identified as a significant problem in theory, as it was pointed out that companies need to understand the benefits of directing their resources and time to collaborative operations to achieve benefits related to utilizing SCC (Småros 2021, 64). As mentioned earlier in this section, the primary driver for companies to utilize SCC and SCI is their potential financial and business advantages. However, this may pose major challenges, as Gutierrez et al. (2020) found that focusing on costs instead of value may create a significant barrier to collaborative efforts in supply chains.

Regarding challenges identified in supply chain integration theory, it was mentioned that terms and incentives related to data sharing should be agreed upon. This may pose challenges for the supply chain members (Ebel et al. 2021) and could create problems in food waste management efforts in the food supply chain between retailers and suppliers. The critical importance of establishing shared objectives, not only related to data sharing but also for the entire collaborative effort among the members, was identified in the empirical results as a means to manage food waste better.

In summary, the relationships between empirical findings and theory were established with respect to both benefits and challenges. It should be noted that empirical findings are difficult to categorize between supply chain collaboration and integration. The empirical study examined these two elements together when exploring the potential benefits and challenges of SCC and SCI in waste management within supply chains.

According to Surucu-Balci and Tuna (2019, 278), food loss and waste is one of the most pressing issues in the world today, given its environmental, social, and economic consequences. The case companies, while noting that food waste management has always been important in their business due to the nature of their products, have also seen an increased focus on FLW management in response to global trends and customer requirements. Furthermore, sustainability and responsibility goals of the case companies promote their efforts towards FLW management in their supply chain operations.

In theory, many solutions directed towards waste management have focused on altering the attitudes and behaviours of individuals (Närvänen et al. 2019, 4). However, the empirical findings of this study suggest that such suboptimization is insufficient when aiming to better manage and mitigate FLW in supply chains. According to the case companies, extensive collaboration between different members of the food supply chain is required to achieve true success in waste management. As Cakar et al. (2020, 1) noted, the wastage of food is a multifaceted issue in food supply chains that expands over various levels and differs between stages. This highlights the need for companies to move away from individual suboptimization and towards a broader understanding of optimization and success in managing and mitigating waste as a whole, encompassing all members of the chain.

These findings are further supported by Strotmann et al. (2017, 2), who noted that solutions to FLW need to be planned and implemented together with other actors and stages. Thus, it is clear that collaboration and cooperation between different members of the food supply chain are crucial to effectively manage and mitigate FLW. The empirical findings of this study suggest that while individual efforts are important, they must be complemented by collective actions to achieve success in waste management.

One of the objectives of the study was to identify best-practices for waste management through the examination of case companies. The empirical findings indicate that open communication, information exchange, and a shared vision among members were identified as critical components of best-practice solutions in utilizing SCC and SCI in waste management. Furthermore, establishing unified goals and forming a clear structure for collaborative supply chain activities, particularly at the retail-supplier interface, was identified as a critical factor in achieving best-practice solutions in the case study. This finding is also supported by Borens et al. (2022), who noted that members should collaborate to establish a baseline and commonly agreed targets for FLW reduction, supported by enablers fostering lasting change.

Additionally, the case companies highlighted that forming a foundation of trust, strong interpersonal relationships, and continuous improvement towards new ideas were essential for achieving best-practice situations. Trust can be noted as a critical best practice element, as Bergman and Holmqvist (2022, 30) pointed out in their research that trust promotes transparency, commitment, and information sharing among the members.

As highlighted by Mathu and Phetla (2018, 2), both supply chain collaboration and integration are indispensable components in achieving supply chain success. It is evident that these elements play a crucial role in the future of waste management in food supply chains, and their importance only increases as markets demand stricter sustainability and responsibility achievements from companies. In the current challenging market environment, where inflation has recently affected customer behaviour and thus impacted the ability of both suppliers and retailers to predict and forecast demand, the focus on waste management through collaborative efforts becomes an even more prominent part, not only in food supply chains but also in other industries.

6 Conclusions

The concluding chapter of this study provides a summary of the main findings. In the second subchapter, the theoretical and managerial contributions of this study are presented. Finally, the last subchapter addresses the study's limitations and proposes potential areas for future research.

6.1 Main findings of the study

This study aimed to evaluate and investigate the use of supply chain collaboration and integration in managing food waste in food supply chains, with a secondary objective of presenting how joint utilization of collaborative practices in FLW management could potentially lead to achieve beneficial outcomes. A main research question and two supporting sub-questions were formulated to achieve these goals.

The study identified SCC and SCI as essential elements for managing FLW in supply chains. The case companies highlighted the importance of effective collaborative communication, aligning incentives through performance indicators, and sharing information regarding forecasting as critical elements for addressing FLW. However, collaborative practices within food supply chain companies have not primarily focused on managing and reducing FLW. Nevertheless, supply chain members are highly interested in minimising and managing FLW. Companies are motivated to use SCC and SCI to achieve benefits such as increased turnover and profit, improved planning and resource management, and increased customer satisfaction resulting from FLW reduction.

The study revealed that SCC and SCI could lead to enhanced supply chain knowledge and a better ability to solve problems within the supply chain, leading to improved business performance. However, constraints and challenges such as insufficient information sharing and resource limitations hinder the use of these collaborative elements. The study identified several factors that food supply chain members should consider when seeking to manage FLW through collaborative supply chain practices, including information sharing, decision

synchronization, incentive alignment, resource sharing, collaborative communication, and joint knowledge creation.

The study found that there is significant room for development in how collaborative supply chain practices are used in FLW management, particularly in retail-supplier partnerships where there is a lack of information sharing and resource sharing. The study also revealed that an integrated approach to SCC and SCI is essential for effective waste management. However, the conceptualization and definition of SCC and SCI pose a challenge. There is a need to form a more precise categorization and identify which elements relate to SCC and which to SCI.

In conclusion, the study underscores the crucial role of SCC and SCI in reducing FLW in supply chains. It also highlights their potential to improve the overall supply chain performance and enhance existing collaboration among supply chain members. The study provides valuable insights for food supply chain managers seeking to leverage SCC and SCI to manage FLW and suggests areas for future research.

6.2 Theoretical and managerial contributions

This study contributes to various research gaps identified by several authors. Firstly, Luo et al. (2022, 16; 18) noted that there has been limited research on food loss and waste reduction, particularly in relation to supply chain cooperation. This study addresses this gap by exploring how collaboration functions in the context of reducing food loss and waste.

Additionally, this study contributes to future research suggestions made by Bergman & Holmqvist (2022, 65–66) who, in their research on Swedish bread suppliers and retailers, pointed out the need for studies to examine how collaboration functions in other food supply chains. Although this study focuses on fresh perishable products, the case study executed involves suppliers from various backgrounds, making it possible to draw broader conclusions about collaborative practices in food supply chains.

Furthermore, this study provides contributions to research on food supply chains, food loss and waste management, and collaborative practices from a qualitative perspective. As noted

by Annosi et al. (2021, 208), previous studies have tended to focus on quantitative analysis, making this study unique in its approach. Also, the study contributes to understanding the dynamics and tensions that arise in collaboration between supply chain members, as highlighted by Annosi et al. (2021, 208). This adds to the body of literature on supply chain cooperation and provides valuable insights for future research.

From a managerial perspective, this study provides valuable insights for food supply chain members on how to achieve better food loss and waste management in their companies and supply chain network. Specifically, the study offers guidance on the collaborative practices that are most promising for FLW management and reduction and provides ideas on how to implement these practices, especially in the retail-supplier interface within food supply chains.

Moreover, this study contributes to a better understanding of the differences between supply chain collaboration and integration, which can provide a significant competitive advantage for companies that can successfully implement these concepts. Therefore, this study can inform supply chain managers and decision-makers on how to improve their supply chain operations by emphasizing and implementing effective collaborative practices.

In addition, from a supplier perspective, the study offers valuable insights into how retailers' approach and view the subject of FLW management and collaborative practices with suppliers. Similarly, from a retailer perspective, the study provides insights into how suppliers view the subject and how they can work together to achieve mutual goals. This knowledge can foster more effective and efficient collaboration between suppliers and retailers in food supply chains.

Overall, the findings of this study have important managerial implications, providing food supply chain members with insights into the most effective collaborative practices for FLW management and reduction, as well as a better understanding of the differences between collaboration and integration. These insights can help companies to achieve a competitive advantage and improve their supply chain operations, ultimately leading to more sustainable and efficient food supply chains. Additionally, this study also makes significant theoretical contributions to food supply chains, food loss and waste management, and supply chain

cooperation. It fills research gaps identified by previous authors and provides insights that can inform future research in these areas.

6.3 Limitations and suggestions for future research

This study also has its limitations that need to be pointed out. Firstly, the study's focus on developed countries and Finnish suppliers and retailers means that the findings may not be applicable to other regions or countries. Therefore, conducting research that examines the topic from the perspectives of suppliers and retailers from different economic and geographical regions would help gain comparative information on how the subject is perceived and acted upon in various areas, even within developed countries.

Another limitation of the study relates to the methodology of the study. The number of companies included in the study, which totalled six, is relatively small compared to the vast number of suppliers and retailers operating in the food industry. Future research could benefit from using a larger sample size to provide more robust findings. Since the study's goal was to understand the underlying nature of the phenomenon and the factors affecting it, qualitative research was fitting in this instance. However, conducting qualitative research with a larger number of case companies could provide valuable insights for further research.

Thirdly, the study only focused on fresh perishable products, which have unique elements and aspects that present distinct requirements when it comes to FLW management and the use of SCC and SCI. While the study's findings can be applied to other product categories, the subject area could also be researched from the perspective of other product groups to identify potential differences. For example, a comparative study could be conducted between fresh products and dry goods. Furthermore, the study's limitation of having only one retailer interviewed could be addressed in future research by involving multiple retailers to gather comparative information. Investigating whether suppliers exhibit different attitudes, policies, and approaches towards the subject with different retailers might provide future research areas.

One suggestion for future research is to separate the concepts of supply chain collaboration and integration. In-depth research that considers supply chain integration in FLW

management could be a potential area for future studies. The present study found that discussing these two elements simultaneously proved challenging, particularly during the empirical phase.

Also, exploring the impact of implementing specific SCC and SCI practices on FLW reduction and management in companies and comparing the results could be a valuable future research subject. Another potential research area could be to compare how SCC and SCI impact companies' FLW amounts and to understand the factors that have led to either a decrease or an increase of FLW in the companies. Such research would require clear and established SCC and SCI practices in place, and a comparison should be made before and after implementing these practices. This kind of research would likely require an extensive amount of data collected over a long period.

The final potential research subject could relate to investigating the role of technology as a part of SCC and SCI in FLW management in food supply chains. Elements of Industry 4.0, such as artificial intelligence, machine learning, data analytics, blockchain technology, and the internet of things, and how these might be used in FLW management in food supply chains, could be a fascinating future research area.

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Appendices

Appendix 1. Interview questions

This interview covers supply chain collaboration, integration as well as food loss and waste (FLW) management topics. This interview focuses on how and why supply chain collaboration and integration can be utilized to manage the amount of FLW generated in food supply chains. This interview aims to highlight the elements related to the retailer and supplier interface. The focus is mainly on the distribution and retail stages of food supply chains; therefore, elements such as primary production and logistics operations are excluded from this study.

BACKGROUND

- Please state your name, the company you are working for and your position in this company.
- Describe in your own words what your company does and in what market the company operate.
- What is the role of supply chain management in your company?

SUPPLY CHAIN COLLABORATION AND INTEGRATION

- Does your company have supply chain collaboration practices in place?
 - If so, what kind of procedures?
- Does your company have supply chain integration practices in place?
 - If so, what kind of procedures?
- How does your company intend to use collaboration and integration in supply chain operations and management?
- Do you see value in supply chain collaboration and integration?

FOOD LOSS AND WASTE MANAGEMENT

- What kind of role do food loss and waste management have in your company?
 - What is your company's viewpoint towards FLW?

- What is the relationship between FLW and the company's supply chain operations and management?
 - Who is responsible for your company's FLW management and development?
- What elements and practices are in place for FLW management?
- How is FLW management reflected in your company's supply chain strategy?
- Is your company actively developing ways to reduce FLW in the food industry, and/or is your company willing to participate actively in developing these?
- Does your company utilize existing data and information in FLW management practices?
- How has the role of FLW management evolved in your company?

SUPPLY CHAIN COLLABORATION AND INTEGRATION IN FOOD LOSS AND WASTE MANAGEMENT

- How has the company implemented supply chain collaboration in regard to FLW management?
 - How about supply chain integration?
- What motivators are enabling supply chain collaboration and integration in FLW management?
- Do you see that supply chain collaboration and/or integration can provide beneficial results in managing FLW?
 - What benefits have you noticed?
- Which factors limit supply chain collaboration and integration in FLW management?
- Has your company used supply chain collaboration and integration with retailers/suppliers to mitigate and manage FLW?
- What challenges may arise from utilizing supply chain collaboration and integration in cooperation with other supply chain members when aiming to manage and mitigate FLW?
- Are there differences between retailers, suppliers, or manufacturers on how they see FLW management in supply chain operations?
 - If so, what kind of differences?
- What role does information exchange have in FLW management?

- What kind of benefits do you see in information exchange? What kind of challenges?
- What are the elements for achieving best practice solutions with other supply chain members in FLW management?
 - What is required to achieve these best practice situations?
- How could supply chain collaboration and integration be developed in terms of FLW management?